Design of a Folded-Cascode CMOS Op Amp

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Abstract

A folded-cascode CMOS op amp is designed in the GF180MCU 3.3 V process [2] using an overdrive-voltage-based sizing method, for a closed-loop application with a feedback factor of 0.5 that must settle to within 2^{-9} in 100 ns. A narrow diode-connected PMOS bias circuit keeps the current sources 150 mV above saturation and helps to increase output swing. NMOS clamps are used to keep the input pair out of the triode region during slewing.

At TT and 27 °C, the amplifier achieves a DC gain of 79.07 dB, a unity-gain frequency of 37.9 MHz, and a phase margin of 82.2°, with a power consumption of 0.418 mW from a 3.3 V supply.

In the closed-loop configuration with $\beta = 0.5$, a rising settling time of 73.5 ns and falling settling time of 70.5 ns are achieved, meeting the maximum 100 ns settling requirement. The NMOS clamps change the settling times by at most 1.1 ns.

The design meets its gain and bandwidth requirements at every process corner and temperature from −40°C to 125°C and in all 500 trials of both Monte Carlo runs. Across all process corners and temperatures, the phase margin stays between 81.1° and 83.0°. The random input offset has a standard deviation of 7.31 mV and mean value of 0.85 mV.

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1 Introduction

This report describes the design of a folded-cascode op amp in the GF180MCU 3.3 V process that meets DC gain and settling requirements.

The target application is a closed-loop amplifier with a feedback factor of $\beta = 0.5$ and a 1 pF load, which must settle to within $\epsilon = 2^{-9}$ of its final value in 100 ns. These targets set a minimum DC gain of 60.2 dB and a minimum unity-gain frequency of 19.86 MHz (Section 4). As a single-stage amplifier, it is compensated by its load capacitance. Therefore, unlike a two-stage op amp, it is stable without the need for a separate compensation network.

The design is simulated in LTspice using the GF180MCU device models. Monte Carlo analysis is run in ngspice, which reads the statistical sections of the PDK model files.

Section 2 describes the circuit and Sections 3 and 4 derive the transistor sizes and bias currents. Section 5 compares two bias circuits and selects a narrow diode-connected PMOS in order to increase the output swing. Sections 6 and 7 present the frequency response and the large-signal step response, including NMOS clamps that improve the slew rate. Section 8 summarizes the final design, including its closed-loop settling at $\beta = 0.5$, its CMRR and power consumption, and Section 9 examines its performance across process corners, temperature and device mismatch.

2 Circuit Description

A schematic of the folded-cascode op amp is shown in Fig. 1. The input differential pair is implemented using an NMOS pair with M_1 and M_2. M_3 and M_4 implement the cascode transistors. M_5, M_6, M_7, and M_8 implement the cascode current mirror. M_9 and M_{10} are PMOS current loads and M_{11} is a tail current source. M_9 and M_{10} operate at DC current I_{B}, and M_{11} operates at I. Therefore, M_1 and M_2 operate at $\frac{I}{2}$. M_3, M_4, M_5, M_6, M_7, and M_8 all operate at $I_{\mathrm{B}} - \frac{I}{2}$. Capacitor C_{L} represents the total output capacitance. V_{B1}, V_{B2}, and V_{B3} are DC bias voltages generated by a separate bias circuit.

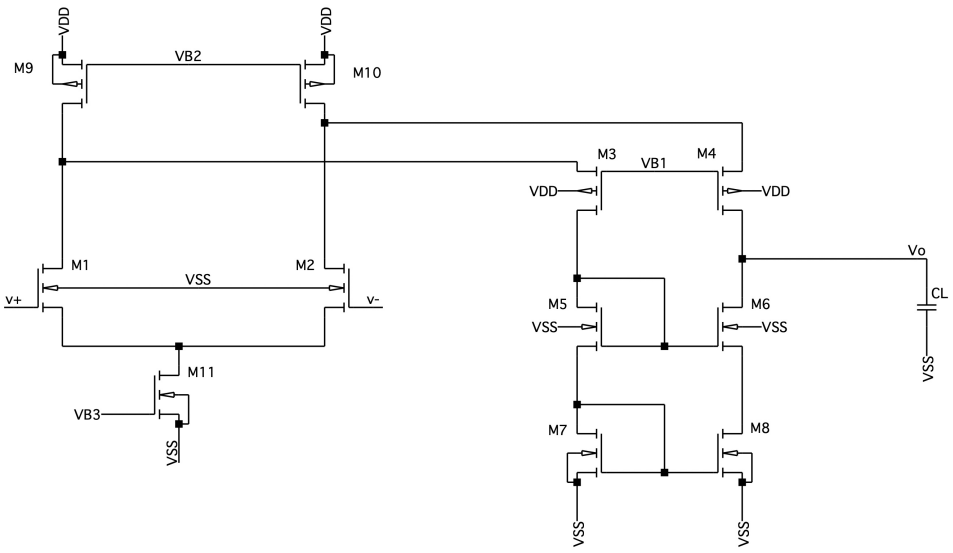


Figure 1: Folded-Cascode CMOS Op Amp Schematic

3 Design Methodology

For the design, a common overdrive voltage $V_{\mathrm{OV}} = 0.2\,\mathrm{V}$ is used for all transistors. As a result, using the square-law relationship, each transistor's DC current is proportional to its W/L. We target $\beta = 0.5\frac{\mathrm{V}}{\mathrm{V}}$ and $\epsilon = 2^{-9}$.

4 DC Bias Point Design

4.1 Requirements and Bias Current

From $\beta = 0.5\,\frac{V}{V}$ and $\epsilon = 2^{-9}$, we have that for the DC gain,

$$A_0 > \frac{1}{\epsilon\,\beta} = \frac{1}{2^{-9}} \cdot \frac{1}{0.5} = 1024\,\mathrm{V/V} = 60.2\,\mathrm{dB}$$

The gain-bandwidth product must be

$$\omega_{\mathrm{t}} > \frac{-\ln(\epsilon)}{\beta\,t_{\mathrm{settle}}} = \frac{-\ln(2^{-9})}{(0.5)(100\times10^{-9}\,\mathrm{s})} = 1.248\times10^{8}\,\mathrm{rad/s} \;\Rightarrow\; f_{\mathrm{t}} = \mathrm{GBW} > \frac{\omega_{\mathrm{t}}}{2\pi} = 19.86\,\mathrm{MHz}$$

The frequency response of the folded-cascode can be approximated as a single-pole system [1]:

$$\frac{v_o}{v_i}(s) = \frac{G_m R_o}{1 + sC_L R_o}$$

Thus,

$$f_{\mathrm{p}} = \frac{1}{2\pi}\,\frac{1}{C_L R_o}$$

and the DC gain is

$$A_0 = G_m R_o$$

Thus,

$$f_{\mathrm{t}} = \frac{1}{2\pi}\,\frac{G_m}{C_L}$$

Thus,

$$2\pi f_{\mathrm{t}}\,C_L = G_m, \quad \text{where } G_m = g_{\mathrm{m1,2}}$$

Also,

$$G_m = g_{\mathrm{m1,2}} = \frac{2\,(I/2)}{V_{\mathrm{OV}}} = \frac{I}{V_{\mathrm{OV}}}$$

We model the output capacitance as $C_L = 1\,\mathrm{pF}$. Thus,

$$G_m = g_{\mathrm{m1,2}} = 2\pi\,(19.86\times10^{6})(10^{-12}) = 0.125\,\mathrm{mS}$$

Thus,

$$I = G_m V_{\mathrm{OV}} = (0.125\,\mathrm{mS})(0.2\,\mathrm{V}) = 25\,\mu\mathrm{A}$$

Thus, the minimum value for I is $25\,\mu$A. We target $I = 60\,\mu$A. Thus, $I_{11} = 60\,\mu$A, $I_{1,2} = 30\,\mu$A, $I_{\mathrm{B}} = 45\,\mu$A, and $I_{3,4,5,6,7,8} = 15\,\mu$A. With $I = 60\,\mu$A, $g_{\mathrm{m1,2}} = 0.3\,\mathrm{mS}$.

4.2 Transistor Sizing

PMOS Sizing Using $V_{\mathrm{SG}} = 0.9\,\mathrm{V}$, and $V_{SD} = V_{DD}/3 = 1.1\,\mathrm{V}$, we perform a DC operating point simulation for the PMOS with $L = 0.56\,\mu\mathrm{m}$ and $W = 1\,\mu\mathrm{m}$. From the simulation, we have that $|V_{\mathrm{th}}| = 0.776\,\mathrm{V}$ and $|V_{\mathrm{dsat}}| = 0.146\,\mathrm{V}$. Determining the V_{SG} at which $V_{\mathrm{OV}} = 0.2\,\mathrm{V}$, we have

$$|V_{\mathrm{OV}}| = V_{\mathrm{SG}} - |V_{\mathrm{th}}| \;\Rightarrow\; V_{\mathrm{SG}} = |V_{\mathrm{OV}}| + |V_{\mathrm{th}}| = 0.2\,\mathrm{V} + 0.776\,\mathrm{V} = 0.976\,\mathrm{V}.$$

Using $V_{\mathrm{SG}} = 0.976\,\mathrm{V}$ and $V_{\mathrm{SD}} = 1.1\,\mathrm{V}$, we have that $|V_{\mathrm{th}}| = 0.776\,\mathrm{V}$ and $|V_{\mathrm{dsat}}| = 0.203\,\mathrm{V}$, which is very close to the target $|V_{\mathrm{OV}}|$ of $0.2\,\mathrm{V}$. At $|V_{\mathrm{SG}}| = 0.976\,\mathrm{V}$ and $|V_{\mathrm{SD}}| = 1.1\,\mathrm{V}$, $|I_{\mathrm{D}}| = 1.46\,\mu\mathrm{A}$. Thus,

$$\frac{I_{\mathrm{D}}}{W} = \frac{1.46\,\mu\mathrm{A}}{1\,\mu\mathrm{m}} = 1.46\,\frac{\mu\mathrm{A}}{\mu\mathrm{m}}$$

NMOS Sizing Using $V_{\mathrm{GS}} = 0.9\,\mathrm{V}$ and $V_{DS} = V_{DD}/3 = 1.1\,\mathrm{V}$, we perform a DC operating point simulation for the NMOS with $L = 0.56\,\mu\mathrm{m}$ and $W = 1\,\mu\mathrm{m}$. From the simulation, we have that $|V_{\mathrm{th}}| = 0.691\,\mathrm{V}$ and $|V_{\mathrm{dsat}}| = 0.223\,\mathrm{V}$. Determining the V_{GS} at which $V_{\mathrm{OV}} = 0.2\,\mathrm{V}$, we have

$$|V_{\mathrm{OV}}| = V_{\mathrm{GS}} - |V_{\mathrm{th}}| \;\Rightarrow\; V_{\mathrm{GS}} = |V_{\mathrm{OV}}| + |V_{\mathrm{th}}| = 0.2\,\mathrm{V} + 0.691\,\mathrm{V} = 0.891\,\mathrm{V}$$

Using $V_{\mathrm{GS}} = 0.891\,\mathrm{V}$ and $V_{\mathrm{DS}} = 1.1\,\mathrm{V}$, we have that $|V_{\mathrm{th}}| = 0.691\,\mathrm{V}$ and $|V_{\mathrm{dsat}}| = 0.216\,\mathrm{V}$, sufficiently close to the target V_{OV} of $0.2\,\mathrm{V}$. At $V_{\mathrm{GS}} = 0.891\,\mathrm{V}$ and $V_{\mathrm{DS}} = 1.1\,\mathrm{V}$, $I_{\mathrm{D}} = 5.95\,\mu\mathrm{A}$. Thus,

$$\frac{I_{\mathrm{D}}}{W} = \frac{5.95\,\mu\mathrm{A}}{1\,\mu\mathrm{m}} = 5.95\,\frac{\mu\mathrm{A}}{\mu\mathrm{m}}$$

Sizing M_1 and M_2 For the NMOS devices, since $I_{\mathrm{M1}} = I_{\mathrm{M2}} = 30\,\mu\mathrm{A}$, using $\frac{I_{\mathrm{D}}}{W} = 5.95\,\mu\mathrm{A}/\mu\mathrm{m}$, we have

$$W_1 = W_2 = \frac{I_{\mathrm{D}}}{5.95\,\mu\mathrm{A}/\mu\mathrm{m}} = \frac{30\,\mu\mathrm{A}}{5.95\,\mu\mathrm{A}/\mu\mathrm{m}} = 5.04\,\mu\mathrm{m}$$

Sizing M_3 and M_4 For the PMOS devices, since $I_{\mathrm{M3}} = I_{\mathrm{M4}} = 15\,\mu\mathrm{A}$, using $\frac{I_{\mathrm{D}}}{W} = 1.46\,\mu\mathrm{A}/\mu\mathrm{m}$, we have

$$W_3 = W_4 = \frac{I_{\mathrm{D}}}{1.46\,\mu\mathrm{A}/\mu\mathrm{m}} = \frac{15\,\mu\mathrm{A}}{1.46\,\mu\mathrm{A}/\mu\mathrm{m}} = 10.27\,\mu\mathrm{m}$$

Sizing M_5, M_6, M_7, M_8 For the NMOS devices, since $I_{\mathrm{M5,M6,M7,M8}} = 15\,\mu\mathrm{A}$, using $\frac{I_{\mathrm{D}}}{W} = 5.95\,\mu\mathrm{A}/\mu\mathrm{m}$, we have

$$W_{5,6,7,8} = \frac{I_{\mathrm{D}}}{5.95\,\mu\mathrm{A}/\mu\mathrm{m}} = \frac{15\,\mu\mathrm{A}}{5.95\,\mu\mathrm{A}/\mu\mathrm{m}} = 2.52\,\mu\mathrm{m}$$

Sizing M_9 and M_{10} For the PMOS devices, since $I_{\mathrm{M9}} = I_{\mathrm{M10}} = I_{\mathrm{B}} = 45\,\mu\mathrm{A}$, using $\frac{I_{\mathrm{D}}}{W} = 1.46\,\mu\mathrm{A}/\mu\mathrm{m}$, we have

$$W_9 = W_{10} = \frac{I_{\mathrm{D}}}{1.46\,\mu\mathrm{A}/\mu\mathrm{m}} = \frac{45\,\mu\mathrm{A}}{1.46\,\mu\mathrm{A}/\mu\mathrm{m}} = 30.82\,\mu\mathrm{m}$$

Sizing M_{11} For the NMOS device, since $I_{\mathrm{M11}} = I = 60\,\mu\mathrm{A}$, using $\frac{I_{\mathrm{D}}}{W} = 5.95\,\mu\mathrm{A}/\mu\mathrm{m}$, we have

$$W_{11} = \frac{I_{\mathrm{D}}}{5.95\,\mu\mathrm{A}/\mu\mathrm{m}} = \frac{60\,\mu\mathrm{A}}{5.95\,\mu\mathrm{A}/\mu\mathrm{m}} = 10.08\,\mu\mathrm{m}$$

4.3 Input Common-Mode Range and Output Swing

Input Common-Mode Range From M_{11}, we have

$$V_{\mathrm{DS11}} \geq V_{\mathrm{OV11}} \Rightarrow\ V_{\mathrm{D11}} - V_{\mathrm{S11}} \geq V_{\mathrm{OV11}}$$

and

$$V_{\mathrm{D11}} = V_{\mathrm{S1,2}} \quad \text{and} \quad V_{\mathrm{GS1,2}} = V_{\mathrm{ICM}} - V_{\mathrm{S1,2}} = V_{\mathrm{OV1,2}} + V_{\mathrm{tn1,2}}$$

$$V_{\mathrm{S1,2}} = V_{\mathrm{ICM}} - V_{\mathrm{OV1,2}} - V_{\mathrm{tn1,2}}$$

$$V_{\mathrm{S11}} = -V_{\mathrm{SS}}$$

Thus,

$$V_{\mathrm{ICM}} - V_{\mathrm{OV1,2}} - V_{\mathrm{tn1,2}} + V_{\mathrm{SS}} \geq V_{\mathrm{OV11}}$$

Thus,

$$\Rightarrow\ V_{\mathrm{ICM}} \geq V_{\mathrm{OV1,2}} + V_{\mathrm{tn1,2}} - V_{\mathrm{SS}} + V_{\mathrm{OV11}}$$

From $M_{9,10}$, we have

$$V_{\mathrm{SD9,10}} \geq V_{\mathrm{OV9,10}}$$

At the edge of saturation, we have $V_{\mathrm{SD9,10}} = V_{\mathrm{OV9,10}}$. Thus,

$$V_{\mathrm{DD}} - V_{\mathrm{D9,10}} = V_{\mathrm{OV9,10}} \ \Rightarrow\ V_{\mathrm{DD}} - V_{\mathrm{OV9,10}} = V_{\mathrm{D9,10}}$$

From $M_{1,2}$, we have

$$V_{\mathrm{GD1,2}} \leq V_{\mathrm{tn1,2}} \ \Rightarrow\ V_{\mathrm{ICM}} - V_{\mathrm{D1,2}} \leq V_{\mathrm{tn1,2}}$$

and $V_{\mathrm{D1,2}} = V_{\mathrm{D9,10}}$. Thus,

$$V_{\mathrm{ICM}} \leq V_{\mathrm{tn1,2}} + V_{\mathrm{D9,10}}$$
$$V_{\mathrm{ICM}} \leq V_{\mathrm{tn1,2}} + V_{\mathrm{DD}} - V_{\mathrm{OV9,10}}$$

Thus,

$$\boxed{V_{\mathrm{OV11}} + V_{\mathrm{OV1,2}} + V_{\mathrm{tn1,2}} - V_{\mathrm{SS}} \ \leq\ V_{\mathrm{ICM}} \ \leq\ V_{\mathrm{tn1,2}} + V_{\mathrm{DD}} - V_{\mathrm{OV9,10}}}$$

Output Swing From $M_{5,6}$, we have

$$V_{\mathrm{DS5,6}} \geq V_{\mathrm{OV5,6}}$$

Thus,

$$V_{\mathrm{o}} - V_{\mathrm{S5,6}} \geq V_{\mathrm{OV5,6}}$$

and

$$V_{\mathrm{S5,6}} = V_{\mathrm{G7,8}} = V_{\mathrm{tn7,8}} + V_{\mathrm{OV7,8}} - V_{\mathrm{SS}}$$

Thus,

$$V_{\mathrm{o}} \geq V_{\mathrm{OV5,6}} + V_{\mathrm{OV7,8}} + V_{\mathrm{tn7,8}} - V_{\mathrm{SS}}$$

From $M_{3,4}$, we have

$$V_{\mathrm{SD3,4}} \geq V_{\mathrm{OV3,4}}$$

Thus,

$$V_{\mathrm{S3,4}} - V_{\mathrm{o}} \geq V_{\mathrm{OV3,4}} \ \Rightarrow\ V_{\mathrm{S3,4}} - V_{\mathrm{OV3,4}} \geq V_{\mathrm{o}}$$

and $V_{\mathrm{S3,4}} = V_{\mathrm{D9,10}}$, and

$$V_{\mathrm{SD9,10}} \geq V_{\mathrm{OV9,10}} \ \Rightarrow\ V_{\mathrm{DD}} - V_{\mathrm{OV9,10}} \geq V_{\mathrm{D9,10}}$$

For maximum output swing, we must have $V_{\text{D9,10}} = V_{\text{DD}} - V_{\text{OV9,10}}$, which is when M_9 and M_{10} are at the edge of saturation. Thus,

$$V_{\text{DD}} - V_{\text{OV9,10}} - V_{\text{OV3,4}} \geq V_{\text{o}}$$

Thus

$$\boxed{V_{\text{OV5,6}} + V_{\text{OV7,8}} + V_{\text{tn7,8}} - V_{\text{SS}} \ \leq \ V_{\text{o}} \ \leq \ V_{\text{DD}} - V_{\text{OV9,10}} - V_{\text{OV3,4}}}$$

If M_9 and M_{10} are not at the edge of saturation, then

$$V_{\text{O,max}} = V_{\text{S3,4}} - V_{\text{OV3,4}}$$

5 Bias Circuit Design

To generate $V_{\text{B1}}, V_{\text{B2}}, V_{\text{B3}}$, an additional bias circuit is used. We consider two such designs: two stacked diode-connected PMOS devices and a narrow diode-connected PMOS.

For $M_{9,10}$, we must have

$$V_{\text{SD9,10}} \geq V_{\text{OV9,10}} \ \Rightarrow \ V_{\text{DD}} - V_{\text{D9,10}} \geq V_{\text{OV9,10}}$$

and

$$V_{\text{D9,10}} = V_{\text{S3,4}}$$

and

$$V_{\text{SG3,4}} = V_{\text{OV3,4}} + |V_{\text{tp3,4}}| \ \Rightarrow \ V_{\text{S3,4}} = V_{\text{OV3,4}} + |V_{\text{tp3,4}}| + V_{\text{B1}}$$

Thus,

$$\begin{aligned} & V_{\text{DD}} - V_{\text{OV9,10}} \geq V_{\text{SG3,4}} + V_{\text{B1}} \\ \Rightarrow \quad & V_{\text{DD}} - V_{\text{OV9,10}} - V_{\text{SG3,4}} \geq V_{\text{B1}} \\ \Rightarrow \ & V_{\text{DD}} - V_{\text{OV9,10}} - V_{\text{OV3,4}} - |V_{\text{tp3,4}}| \geq V_{\text{B1}} \end{aligned}$$

Thus, the maximum value for V_{B1} is

$$\boxed{V_{\text{B1,max}} = V_{\text{DD}} - V_{\text{OV9,10}} - V_{\text{OV3,4}} - |V_{\text{tp3,4}}|}$$

This occurs when $M_{9,10}$ are at the edge of saturation, when

$$V_{\text{SD9,10}} = V_{\text{OV9,10}}$$

5.1 Two Stacked Diode-Connected PMOS

An implementation for the bias circuit is shown in Fig. 2.

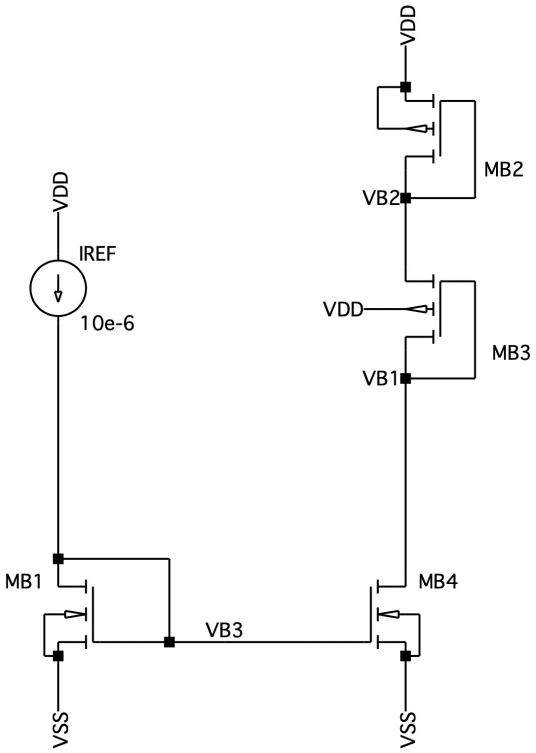


Figure 2: Two Stacked Diode-Connected PMOS Schematic

Determining V_{B1}, we have

$$V_{\mathrm{SGB3}} = V_{\mathrm{B2}} - V_{\mathrm{B1}} = V_{\mathrm{OVB3}} + |V_{\mathrm{tpB3}}|$$

and

$$V_{\mathrm{SGB2}} = V_{\mathrm{DD}} - V_{\mathrm{B2}} = V_{\mathrm{OVB2}} + |V_{\mathrm{tpB2}}|$$

M_{B3} will be affected by the body effect. Thus, we model the threshold voltages as

$$|V_{\mathrm{tpB3}}| = |V_{\mathrm{tp}}| + \Delta V_{\mathrm{t}}, \qquad |V_{\mathrm{tpB2}}| = |V_{\mathrm{tp}}|$$

Thus, we have

$$V_{\mathrm{DD}} - V_{\mathrm{OVB2}} - |V_{\mathrm{tpB2}}| - V_{\mathrm{OVB3}} - |V_{\mathrm{tpB3}}| = V_{\mathrm{B1}}$$

Since $V_{\mathrm{OVB2}} = V_{\mathrm{OVB3}} = V_{\mathrm{OV}}$,

$$V_{\mathrm{DD}} - 2V_{\mathrm{OV}} - 2|V_{\mathrm{tp}}| - \Delta V_{\mathrm{t}} = V_{\mathrm{B1}}$$

This value is lower than the max value for V_{B1}.

From $M_{9,10}$ we have

$$V_{\mathrm{SD9,10}} = V_{\mathrm{DD}} - V_{\mathrm{D9,10}}$$

where $V_{\mathrm{D9,10}} = V_{\mathrm{S3,4}} = V_{\mathrm{OV3,4}} + |V_{\mathrm{tp3,4}}| + V_{\mathrm{B1}}$. Thus,

$$V_{\mathrm{SD9,10}} = V_{\mathrm{DD}} - V_{\mathrm{OV3,4}} - |V_{\mathrm{tp3,4}}| - V_{\mathrm{B1}}$$

Thus,

$$\begin{aligned} V_{\mathrm{SD9,10}} &= V_{\mathrm{DD}} - V_{\mathrm{OV3,4}} - |V_{\mathrm{tp3,4}}| - [V_{\mathrm{DD}} - V_{\mathrm{OVB2}} - V_{\mathrm{OVB3}} - |V_{\mathrm{tpB2}}| - |V_{\mathrm{tpB3}}|] \\ &= V_{\mathrm{OVB2}} + V_{\mathrm{OVB3}} - V_{\mathrm{OV3,4}} + |V_{\mathrm{tpB2}}| + |V_{\mathrm{tpB3}}| - |V_{\mathrm{tp3,4}}| \end{aligned}$$

Since M_{B3} and $M_{3,4}$ operate at the same current density and with the same body bias, $V_{OVB3} \approx V_{OV3,4}$ and $|V_{tpB3}| \approx |V_{tp3,4}|$. Thus,

$$V_{SD9,10} \approx V_{OVB2} + |V_{tpB2}| = V_{SGB2} > V_{OV9,10}$$

Since $V_{OVB2} = V_{OV9,10}$ by the current mirror, $M_{9,10}$ are in saturation with a margin of approximately $|V_{tpB2}|$. The margin is much higher than needed, which reduces the maximum value for the output.

5.2 Narrow Diode-Connected PMOS

An implementation for the bias circuit is shown in Fig. 3.

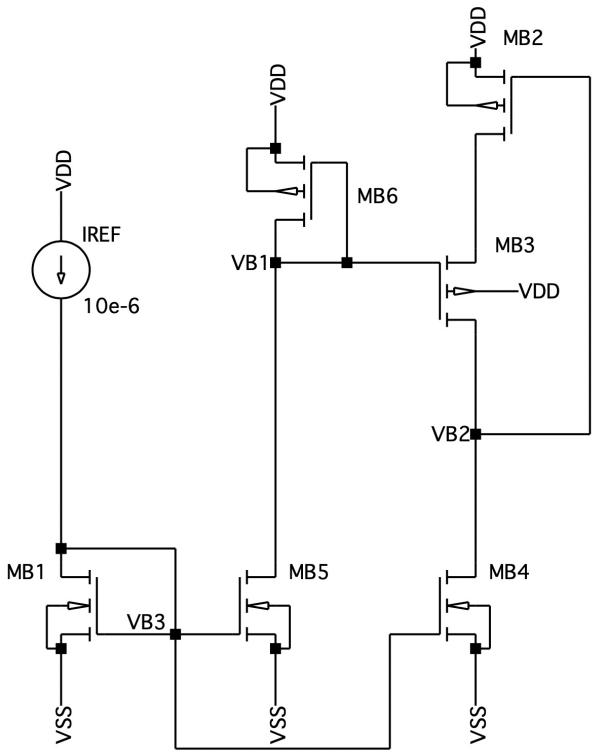


Figure 3: Narrow Diode-Connected PMOS Schematic

We narrow M_{B6} by decreasing its width, compared to the other PMOS in the bias circuit. We have

$$V_{B1} = V_{DD} - |V_{tpB6}| - V_{OVB6}$$

We also have

$$\begin{aligned} V_{SD9,10} &= V_{DD} - V_{B1} - |V_{tp3,4}| - V_{OV3,4} \\ &= V_{DD} - [V_{DD} - |V_{tpB6}| - V_{OVB6}] - |V_{tp3,4}| - V_{OV3,4} \\ &= V_{OVB6} - V_{OV3,4} + |V_{tpB6}| - |V_{tp3,4}| \end{aligned}$$

$M_{3,4}$ are affected by the body effect. Thus,

$$|V_{tp3,4}| = |V_{tp}| + \Delta V_t, \qquad |V_{tpB6}| = |V_{tp}|$$

Thus,

$$V_{SD9,10} = V_{OVB6} - V_{OV3,4} + |V_{tp}| - (|V_{tp}| + \Delta V_t) = V_{OVB6} - V_{OV3,4} - \Delta V_t$$

Since $I_{B6} = I_{B2}$ and the two devices have the same length,

$$\sqrt{W_{B6}}\, V_{OVB6} = \sqrt{W_{B2}}\, V_{OVB2}$$

We make M_{B6} narrower than M_{B2}, with

$$W_{B6} = \frac{1}{n} W_{B2}$$

Since M_{B2}, $M_{3,4}$, and $M_{9,10}$ have the same current density,

$$V_{OVB2} = V_{OV3,4} = V_{OV9,10}$$

Thus,

$$V_{OVB6} = \sqrt{n}\, V_{OVB2} = \sqrt{n}\, V_{OV3,4}$$

and

$$V_{SD9,10} = \sqrt{n}\, V_{OV3,4} - V_{OV3,4} - \Delta V_t = \left(\sqrt{n} - 1\right) V_{OV3,4} - \Delta V_t$$

We choose a margin of 0.150 V above $V_{OV9,10}$ for $M_{9,10}$, so that $V_{SD9,10} = 0.35\,\text{V}$. Thus, $V_{D9,10} = V_{S3,4} = 2.95\,\text{V}$. We simulate ΔV_t for $M_{3,4}$ using a diode-connected PMOS with its body connected to V_{DD} and then $V_{D9,10}$. The source is connected to 2.95 V. A schematic of this is shown in Fig. 4.

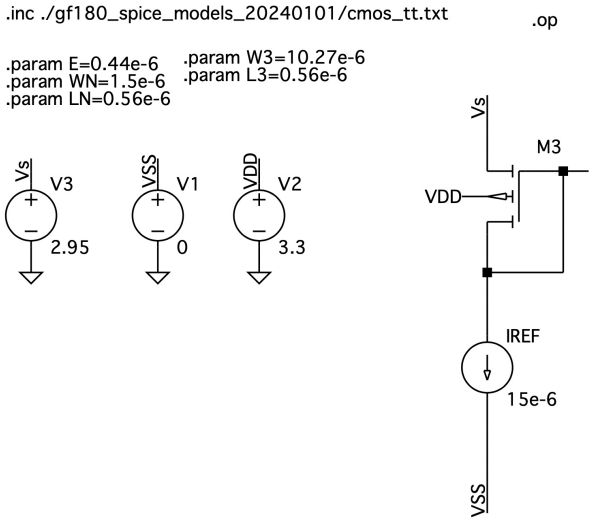


Figure 4: Diode-Connected PMOS Schematic

Simulating with the body at V_{DD} and then at 2.95 V, we have

$$\Delta V_t = 0.935\,\text{V} - 0.790\,\text{V} = 0.145\,\text{V}$$

Thus,

$$V_{SD9,10} = \left(\sqrt{n} - 1\right) V_{OV3,4} - \Delta V_t = V_{OV3,4} + m$$

where $m = 0.150\,\text{V}$ is the margin. Thus,

$$\sqrt{n} = \frac{m + \Delta V_t}{V_{OV3,4}} + 2 = \frac{0.150\,\text{V} + 0.145\,\text{V}}{0.2\,\text{V}} + 2 = 3.475 \;\Rightarrow\; n \approx 12$$

Thus,

$$W_{B6} = \frac{1}{12} W_{B2} = \frac{6.85\,\mu\text{m}}{12} = 0.57\,\mu\text{m}$$

Tuning the bias circuit in simulation, $n = 9.8$ ($W_{B6} = 0.7\,\mu\text{m}$) provides the desired margin of 150 mV for $M_{9,10}$, with $V_{SD9,10} = 0.358\,\text{V}$. The difference from the calculated value is due to the deviation from square-law behaviour.

5.3 Sizing Narrow Diode Bias Circuit

Sizing M_{B1}, M_{B4}, and M_{B5} For the NMOS devices, since $I_{\mathrm{B1,B4,B5}} = I_{REF} = 10\,\mu\mathrm{A}$, using $\frac{I_{\mathrm{D}}}{W} = 5.95\,\mu\mathrm{A}/\mu\mathrm{m}$, we have

$$W_{\mathrm{B1,B4,B5}} = \frac{I_{\mathrm{D}}}{5.95\,\mu\mathrm{A}/\mu\mathrm{m}} = \frac{10\,\mu\mathrm{A}}{5.95\,\mu\mathrm{A}/\mu\mathrm{m}} = 1.68\,\mu\mathrm{m}$$

Sizing M_{B2} and M_{B3} For the PMOS devices, since $I_{\mathrm{B2,B3}} = I_{REF} = 10\,\mu\mathrm{A}$, using $\frac{I_{\mathrm{D}}}{W} = 1.46\,\mu\mathrm{A}/\mu\mathrm{m}$, we have

$$W_{\mathrm{B2,B3}} = \frac{I_{\mathrm{D}}}{1.46\,\mu\mathrm{A}/\mu\mathrm{m}} = \frac{10\,\mu\mathrm{A}}{1.46\,\mu\mathrm{A}/\mu\mathrm{m}} = 6.85\,\mu\mathrm{m}$$

5.4 DC Operating Point

Figure 5 shows the full schematic for the circuit, with the bias circuit implemented using the narrow diode circuit.

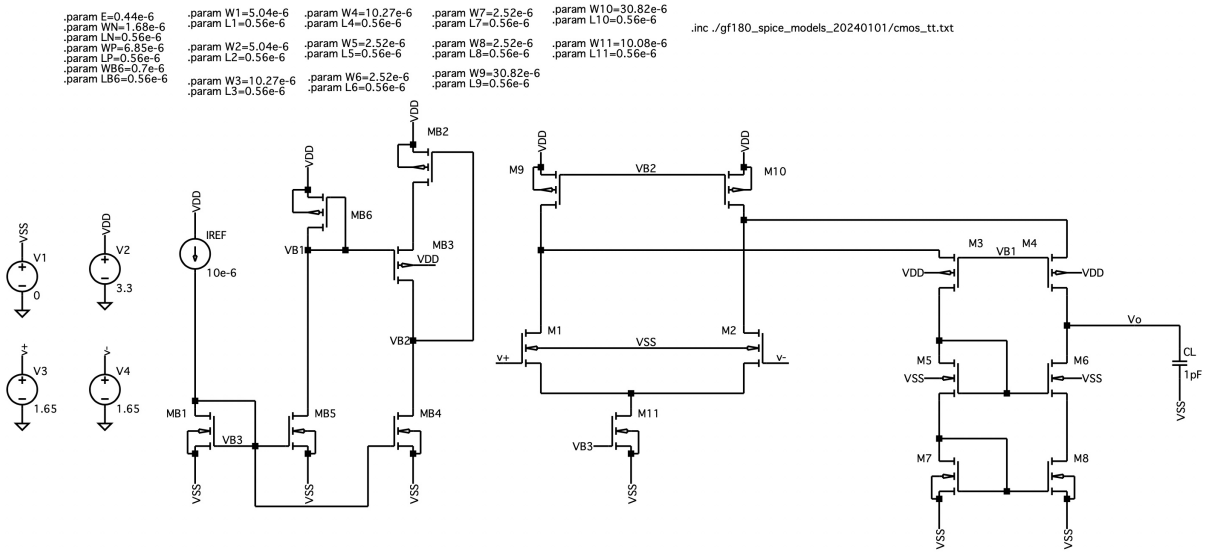


Figure 5: Full Folded-Cascode Schematic

Table 1 summarizes the sizing for the transistors.

Table 1: Transistor Sizing Summary

Transistor	Type	Width (μm)	Length (μm)
M1	NMOS	5.04	0.56
M2	NMOS	5.04	0.56
M3	PMOS	10.27	0.56
M4	PMOS	10.27	0.56
M5	NMOS	2.52	0.56
M6	NMOS	2.52	0.56
M7	NMOS	2.52	0.56
M8	NMOS	2.52	0.56
M9	PMOS	30.82	0.56
M10	PMOS	30.82	0.56
M11	NMOS	10.08	0.56
MB1	NMOS	1.68	0.56
MB2	PMOS	6.85	0.56
MB3	PMOS	6.85	0.56
MB4	NMOS	1.68	0.56
MB5	NMOS	1.68	0.56
MB6	PMOS	0.7	0.56

A summary of the DC operating point data is provided in Tables 2 to 4:

Table 2: DC Operating Point of the Amplifier Devices

Transistor	$M_{1,2}$	$M_{3,4}$	$M_{5,6}$	$M_{7,8}$	$M_{9,10}$	M_{11}
I_{D} (μA)	30.0	17.7	17.7	17.7	47.7	60.0
$\lvert V_{\mathrm{GS}}\rvert$ (V)	1.07	1.15	1.18	0.912	0.992	0.892
$\lvert V_{\mathrm{DS}}\rvert$ (V)	2.36	0.848	1.18	0.912	0.359	0.583
$\lvert V_{\mathrm{BS}}\rvert$ (V)	0.583	0.359	0.912	0	0	0
$\lvert V_{\mathrm{th}}\rvert$ (V)	0.887	0.938	0.970	0.703	0.790	0.708
$\lvert V_{\mathrm{dsat}}\rvert$ (V)	0.208	0.220	0.236	0.224	0.202	0.205
g_{m} (μS)	251	142	135	134	396	495
g_{mb} (μS)	69.4	53.6	33.4	48.8	179	180
g_{ds} (μS)	1.06	0.847	0.774	0.940	5.71	4.27
C_{bd} (fF)	3.87	8.62	2.14	2.49	29.4	10.2
C_{bs} (fF)	5.16	9.87	2.49	3.12	32.4	12.1

Table 3: DC Operating Point of the Bias Devices

Transistor	M_{B1}	M_{B2}	M_{B3}	M_{B4}	M_{B5}	M_{B6}
I_{D} (μA)	10.0	10.6	10.6	10.6	10.4	10.4
$\lvert V_{\mathrm{GS}}\rvert$ (V)	0.892	0.992	1.14	0.892	0.892	1.51
$\lvert V_{\mathrm{DS}}\rvert$ (V)	0.892	0.367	0.625	2.31	1.79	1.51
$\lvert V_{\mathrm{BS}}\rvert$ (V)	0	0	0.367	0	0	0
$\lvert V_{\mathrm{th}}\rvert$ (V)	0.699	0.789	0.941	0.698	0.698	0.771
$\lvert V_{\mathrm{dsat}}\rvert$ (V)	0.211	0.203	0.212	0.212	0.212	0.617
g_{m} (μS)	80.9	88.0	89.1	84.6	83.5	25.2
g_{mb} (μS)	29.6	39.9	33.4	31.0	30.5	11.1
g_{ds} (μS)	0.583	1.22	0.654	0.397	0.424	0.357
C_{bd} (fF)	1.71	6.61	5.95	1.44	1.51	0.686
C_{bs} (fF)	2.13	7.31	6.61	2.13	2.13	0.866

Table 4: DC Node Voltages and Power Consumption

Quantity	Value
V_{B1}	1.792 V
V_{B2}	2.308 V
V_{B3}	0.892 V
Folding Nodes ($V_{\mathrm{D9,10}}$)	2.941 V
Tail Node (V_{D11})	0.583 V
Output (V_{O})	2.093 V
Total Supply Current	126.6 μA
Power Consumption	0.418 mW

Determining the input common-mode range and output swing, we have

$$V_{\mathrm{ICM,min}} = V_{\mathrm{OV11}} + V_{\mathrm{OV1,2}} + V_{\mathrm{tn1,2}} - V_{\mathrm{SS}} = 1.3\,\mathrm{V}$$
$$V_{\mathrm{ICM,max}} = V_{\mathrm{tn1,2}} + V_{\mathrm{DD}} - V_{\mathrm{OV9,10}} = 3.985\,\mathrm{V}$$

$$V_{\mathrm{O,min}} = V_{\mathrm{OV5,6}} + V_{\mathrm{OV7,8}} + V_{\mathrm{tn7,8}} - V_{\mathrm{SS}} = 1.163\,\mathrm{V}$$
$$V_{\mathrm{O,max}} = V_{\mathrm{S3,4}} - V_{\mathrm{OV3,4}} = V_{B1} + |V_{\mathrm{tp3,4}}| = 2.73\,\mathrm{V}$$

$$V_{\mathrm{B1,max}} = V_{\mathrm{DD}} - V_{\mathrm{OV9,10}} - V_{\mathrm{OV3,4}} - |V_{\mathrm{tp3,4}}| = 1.939\,\mathrm{V}$$

We see that $V_{\mathrm{ICM,max}} = 3.985\,\mathrm{V} > V_{\mathrm{DD}} = 3.3\,\mathrm{V}$.

6 Small-Signal AC Analysis

6.1 Hand Analysis

For the folded-cascode, the DC gain is [1]

$$A_0 = g_{\mathrm{m1,2}} \left\{ \left[\left(g_{\mathrm{m3,4}} + g_{\mathrm{mb3,4}} \right) r_{\mathrm{o3,4}} \left(r_{\mathrm{o1,2}} \parallel r_{\mathrm{o9,10}} \right) \right] \parallel \left[\left(g_{\mathrm{m5,6}} + g_{\mathrm{mb5,6}} \right) r_{\mathrm{o5,6}} \, r_{\mathrm{o7,8}} \right] \right\}$$

With $r_{\mathrm{o}} = 1/g_{\mathrm{ds}}$, we have

$$r_{\mathrm{o1,2}} = 943\,\mathrm{k\Omega}, \qquad r_{\mathrm{o3,4}} = 1.18\,\mathrm{M\Omega}, \qquad r_{\mathrm{o5,6}} = 1.29\,\mathrm{M\Omega},$$
$$r_{\mathrm{o7,8}} = 1.06\,\mathrm{M\Omega}, \qquad r_{\mathrm{o9,10}} = 175\,\mathrm{k\Omega}$$

Thus,

$$\begin{aligned} A_0 &= g_{\mathrm{m1,2}} \left\{ \left[\left(g_{\mathrm{m3,4}} + g_{\mathrm{mb3,4}} \right) r_{\mathrm{o3,4}} \left(r_{\mathrm{o1,2}} \parallel r_{\mathrm{o9,10}} \right) \right] \parallel \left[\left(g_{\mathrm{m5,6}} + g_{\mathrm{mb5,6}} \right) r_{\mathrm{o5,6}} \, r_{\mathrm{o7,8}} \right] \right\} \\ &= (251\,\mu\mathrm{S}) \Big\{ \left[(142\,\mu\mathrm{S} + 53.6\,\mu\mathrm{S})(1.18\,\mathrm{M\Omega})(943\,\mathrm{k\Omega} \parallel 175\,\mathrm{k\Omega}) \right] \\ &\qquad \parallel \left[(135\,\mu\mathrm{S} + 33.4\,\mu\mathrm{S})(1.29\,\mathrm{M\Omega})(1.06\,\mathrm{M\Omega}) \right] \Big\} \\ &= (251\,\mu\mathrm{S}) \left\{ 34.1\,\mathrm{M\Omega} \parallel 231.5\,\mathrm{M\Omega} \right\} \\ &= (251\,\mu\mathrm{S})(29.7\,\mathrm{M\Omega}) \\ &= 7460\,\mathrm{V/V} = 77.5\,\mathrm{dB} \end{aligned}$$

6.2 Simulated Response

Figure 6 shows the setup for AC analysis, with a load capacitor of 1 pF. The capacitor C_{FB} and inductor L_{FB} are used to set the DC operating point of the op amp. The netlist can be found in Section B.1 with M_{12} and M_{13} removed.

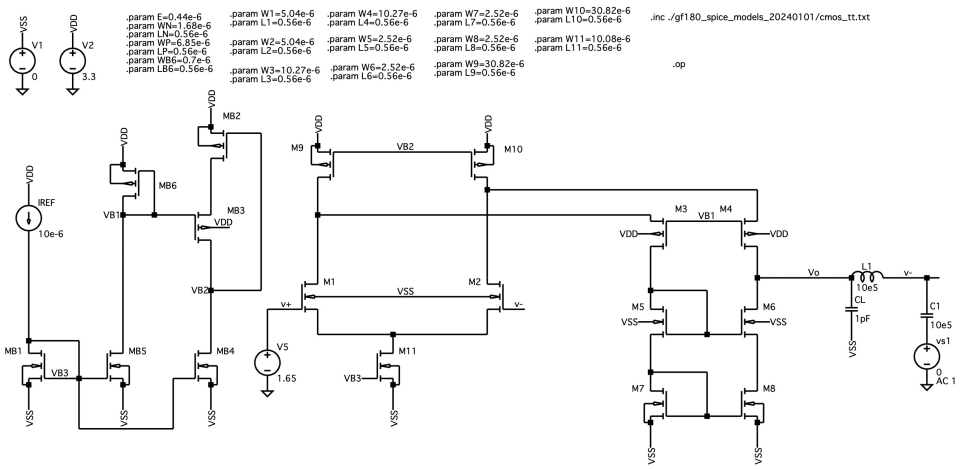


Figure 6: Open-loop Frequency Response Setup

Performing the AC analysis, we have the following frequency response in Fig. 7:

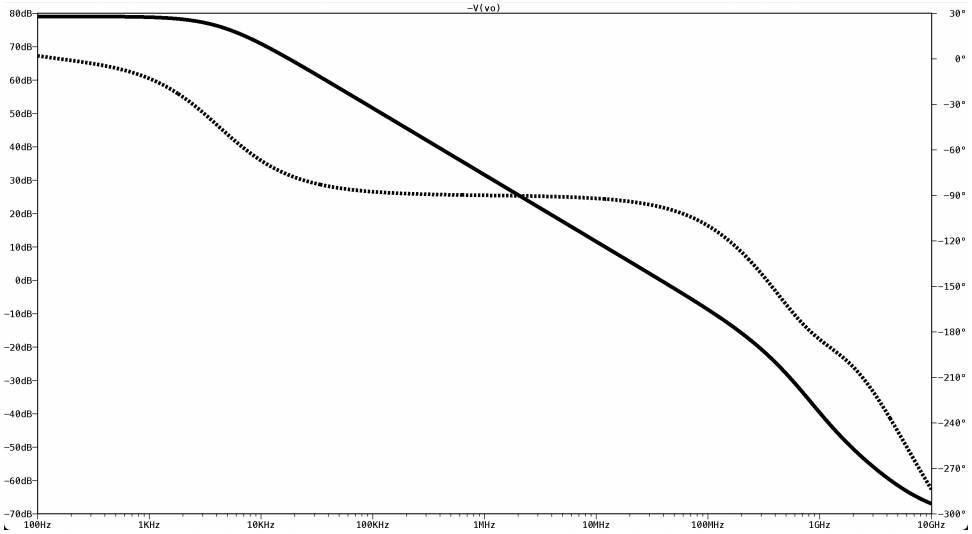


Figure 7: Open-loop Frequency Response

The simulated DC gain is 79.07 dB (8988 V/V). This is 1.6 dB higher than the hand calculation, which used the small-signal parameters of Table 2, taken with the output at 2.09 V. In the AC test bench, the output is held at 1.65 V by the DC feedback through L_{FB}. We find that $f_{\mathrm{t}} = 37.87\,\mathrm{MHz}$, where the phase is $-97.7°$, and the $-3\,\mathrm{dB}$ bandwidth is 4.24 kHz. Therefore, the phase margin is

$$\mathrm{PM} = 180° - 97.7° = 82.3°$$

The phase reaches $-180°$ at 835.7 MHz, giving a gain margin of 36.5 dB.

7 Time-Domain Behaviour and Slew Rate

7.1 Slew Rate

When $V_{\text{id}} = V_{\text{G1}} - V_{\text{G2}}$ is large, M_2 turns off [1]. As a result, M_1 tries to conduct the entire bias current I. This is not possible, because $I > I_{\text{B}}$. At the drain of M_1,

$$I_{\text{B}} = I_{\text{M1}} + I_{\text{M3}} \;\Rightarrow\; I_{\text{M3}} = I_{\text{B}} - I_{\text{M1}}$$

and the minimum value of I_{M3} is $0\,\mu\text{A}$. Therefore, M_1 can conduct at most I_{B}, which is less than the current I it would otherwise carry. As a result, the voltage at the drain of M_1 falls and M_1 enters the triode region. Since M_2 is off, the current in M_{11} must also fall to I_{B}, so M_{11} enters the triode region as well. With $I_{\text{M3}} = 0\,\mu\text{A}$, the current mirror carries no current, so $I_{\text{M5}} = I_{\text{M6}} = 0\,\mu\text{A}$. Since M_2 is off, $I_{\text{M4}} = I_{\text{B}}$. Thus, I_{B} flows into C_{L}, and the slew rate is [1]

$$\text{SR} = \frac{I_{\text{B}}}{C_{\text{L}}}$$

After slewing, additional time is needed for M_1 and M_{11} to leave the triode region and return to saturation. This lengthens the settling time and can distort the output. For the open loop large step response, we set $V_+ = 2\,\text{V}, V_- = 0.9\,\text{V}$. Performing a DC operating point, we have the summary Table 5:

Table 5: Open-Loop Operating Point Without Clamps ($V_+ = 2.0\,\text{V}$, $V_- = 0.9\,\text{V}$)

Transistor	I_{D}	$\lvert V_{\text{DS}}\rvert$ (V)	$\lvert V_{\text{dsat}}\rvert$ (V)	Region
M_{11}	53.8 μA	0.186	0.204	Triode
M_1	52.6 μA	0.047	0.769	Triode
M_2	1.28 μA	3.11	0.066	Off (subthreshold)
M_9	52.6 μA	3.07	0.203	Saturation
M_3	2.9 pA	–	–	Off

We see that M_1 and M_{11} are indeed in the triode region and conduct less than $I = 60\,\mu\text{A}$. M_1 conducts $52.6\,\mu\text{A}$ rather than $I_{\text{B}} = 45\,\mu\text{A}$ as we have ignored channel-length modulation.

7.2 Use of NMOS Clamps to Increase Slew Rate

NMOS clamps can be used to ensure M_1 and M_{11} do not enter the triode region [1]. They also increase the current that charges C_L. A schematic of the folded-cascode with the addition of the NMOS clamps is shown in Fig. 8:

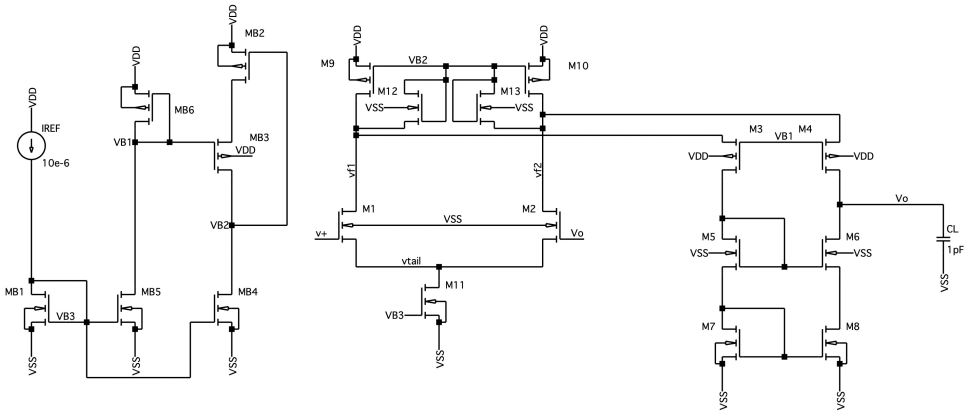


Figure 8: Folded-Cascode Schematic with NMOS Clamps

M_{12} and M_{13} implement the clamps. During normal operation, both clamps will be off. This can be seen since

$$V_{SG3,4} = V_{D1,2} - V_{B1} = V_{OV3,4} + V_{tp3,4} \Rightarrow V_{S12,13} = V_{D1} = V_{OV3,4} + V_{tp3,4} + V_{B1}$$

And

$$V_{B2} = V_{DD} - V_{OVB2} - V_{tpB2}$$
$$V_{B1} = V_{DD} - V_{OVB6} - V_{tpB6}$$

Since $M_{3,4}$ are affected by the body effect,

$$V_{tp3,4} = V_t + \Delta V_t$$

$$V_{tpB6} = V_t$$

And

$$V_{OV3,4} = V_{OV}$$
$$V_{OVB6} = \sqrt{n} V_{OV3,4} = \sqrt{n} V_{OV}$$

Thus,

$$\begin{aligned} V_{D1} &= V_{OV3,4} + |V_{tp3,4}| + V_{B1} \\ &= V_{OV} + [V_t + \Delta V_t] + [V_{DD} - \sqrt{n}\, V_{OV} - V_t] \\ &= V_{DD} + (1 - \sqrt{n})\, V_{OV} + \Delta V_t \approx 3.02\,\mathrm{V} \end{aligned}$$

and

$$V_{B2} = V_{DD} - V_{OVB2} - V_{tpB2} \approx 2.31\,\mathrm{V}$$

Thus,

$$V_{G12,13} - V_{S12,13} = V_{B2} - V_{D1} < 0$$

Thus, $M_{12,13}$ are off. When M_1 goes into the triode region, M_{12} will turn on and conduct. Similarly, when M_2 goes into the triode region, M_{13} will turn on and conduct. In the case when M_1 is on and conducts the entire tail current and M_{12} turns on, the current of M_{12} will cause the current drawn by M_{B2} to increase:

$$I_{MB2} = I_{MB4} + I_{M12} = 10\,\mu\mathrm{A} + I_{M12}$$

Since M_9 and M_{10} form a current mirror with M_{B2}:

$$I_9 = \frac{W_9}{W_{\mathrm{B2}}}\, I_{\mathrm{MB2}} = \frac{30.82\,\mu\mathrm{m}}{6.85\,\mu\mathrm{m}}\,(10\,\mu\mathrm{A} + I_{\mathrm{M12}}) = 4.5\,(10\,\mu\mathrm{A} + I_{\mathrm{M12}})$$

We also have:

$$I_{M9} + I_{M12} = 60\,\mu\mathrm{A}$$

Solving, we have that

$$I_{M9} = 57.3\,\mu\mathrm{A}$$
$$I_{M12} = 2.73\,\mu\mathrm{A}$$
$$I_{MB2} = 12.73\,\mu\mathrm{A}$$

I_{M10} will charge C_L. Therefore, the current to charge C_L increases from $I_{M9,M10} = I_B = 45\,\mu\mathrm{A}$ to $I_{M9,M10} = 57.3\,\mu\mathrm{A}$. Therefore, the slew rate increases by a factor of

$$\frac{57.3\,\mu\mathrm{A}}{45\,\mu\mathrm{A}} = 1.27$$

Sizing M_{12} and M_{13} We let $W_{12,13} = 2\,\mu$m and $L_{12,13} = 0.56\,\mu$m. Performing a DC operating point, we have the summary in Table 6:

Table 6: Open-Loop Operating Point With Clamps ($V_+ = 2.0\,$V, $V_- = 0.9\,$V)

Transistor	I_D	$\lvert V_{DS}\rvert$ (V)	$\lvert V_{dsat}\rvert$ (V)	Region
M_{11}	60.7 μA	0.770	0.205	Saturation
M_1	60.7 μA	0.455	0.298	Saturation
M_2	3.3 pA	–	–	Off
M_9	59.1 μA	2.08	0.216	Saturation
M_{12}	1.58 μA	1.07	0.102	Saturation
M_{B2}	12.2 μA	0.355	0.216	Saturation
M_3	2.9 pA	–	–	Off

We see that I_1 and I_{11} have both increased, and M_1 and M_{11} are now both in saturation.

Performing the large step response with and without clamps, we have the following in Table 7. These step responses use the unity-gain buffer test bench of Section B.2, with a 1 V step from 1.5 to 2.5 V.

Table 7: Unity-Gain Step Response With and Without Clamps ($C_L = 1\,$pF, 1.5 V to 2.5 V Step)

	Without Clamps	With Clamps
Rising Slew Rate (V/μs)	39.3	44.4
Falling Slew Rate (V/μs)	44.6	46.4
Rise Time, 10% to 90% (ns)	20.4	18.0
Fall Time, 90% to 10% (ns)	17.9	17.2
Settling Time to 2^{-9}, Rising (ns)	38.9	36.8
Settling Time to 2^{-9}, Falling (ns)	35.0	30.9
Overshoot (mV)	19.1	19.8
Undershoot (mV)	20.0	10.6
I_{D2} at $V_O = 2.0\,$V (μA)	5.64	0.67

We see that the increase in rising slew rate is

$$\frac{44.4\,\text{V}/\mu s}{39.3\,\text{V}/\mu s} = 1.13$$

This is less than the calculated 1.27 factor. This can be attributed to ignoring channel-length modulation in the calculations and assuming that one input transistor is off for the whole duration of slewing. Settling time is measured as the time for the output to remain within $\pm 1.95\,$mV (2^{-9} of the 1 V step) of the input. Without clamps the amplifier settles in 38.9 ns on the rising edge and 35.0 ns on the falling edge. Both are within the 100 ns requirement. With the clamps, this is reduced to 36.8 ns for the rising edge and 30.9 ns for the falling edge. With both inputs at 1.65 V, the clamp transistors M_{12} and M_{13} have $V_{GS} = -0.633\,$V and $I_{12,13} = 2.3\,$pA. All branch currents and the folding-node voltages (2.94 V) are unchanged from the operating point without clamps. The clamps therefore have no effect on the bias point in normal operation.

Since the clamps are off during normal operation, they contribute no static current. The benefit of the clamps in this design is small because of the small load capacitance of 1 pF.

7.3 Demonstrating Clamp Behaviour

We can demonstrate the effect of the clamps using a 5 pF load. These step responses use the test bench of Section B.2 with $C_{\mathrm{L}} = 5\,\mathrm{pF}$ and a 0.6 V step from 1.4 to 2.0 V. Figures 9 and 10 show the rising and falling edges.

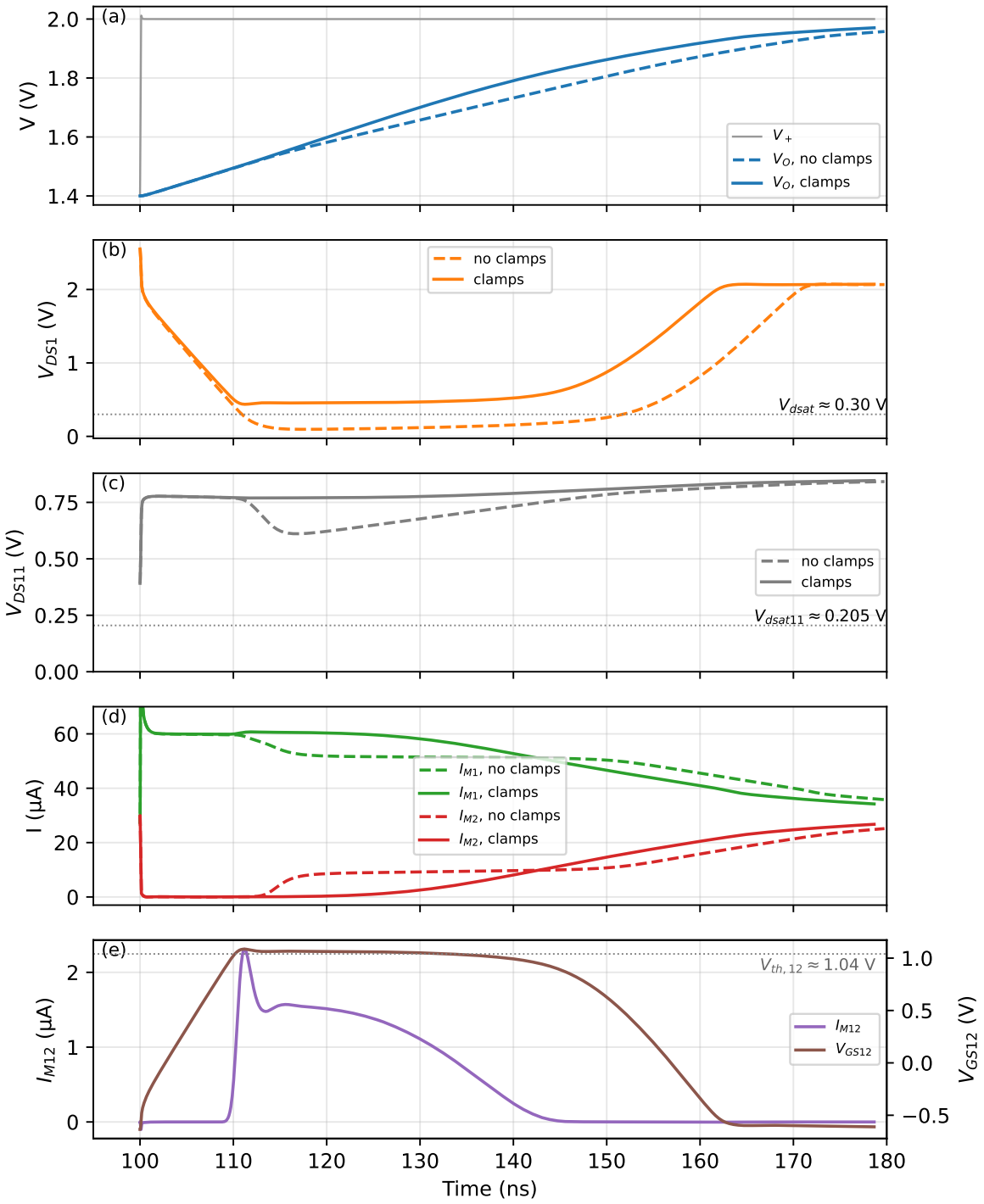


Figure 9: Rising edge of the unity-gain step response with $C_{\mathrm{L}} = 5\,\mathrm{pF}$ (1.4 V to 2.0 V), without clamps (dashed) and with clamps (solid). (a) Input and output. (b) V_{DS1}: without clamps it falls to 96 mV, below V_{dsat}; with clamps it falls to only 0.437 V. (c) V_{DS11}: the tail source stays in saturation in both runs; without clamps it sags to 0.61 V when M_2 turns back on, while with clamps it stays near 0.77 V. (d) Input-pair currents: without clamps M_2 conducts about 9 μA during slewing; with clamps it remains off. (e) Clamp M_{12} turns on at about 110 ns, holding V_{GS12} near its threshold while conducting up to 2.3 μA.

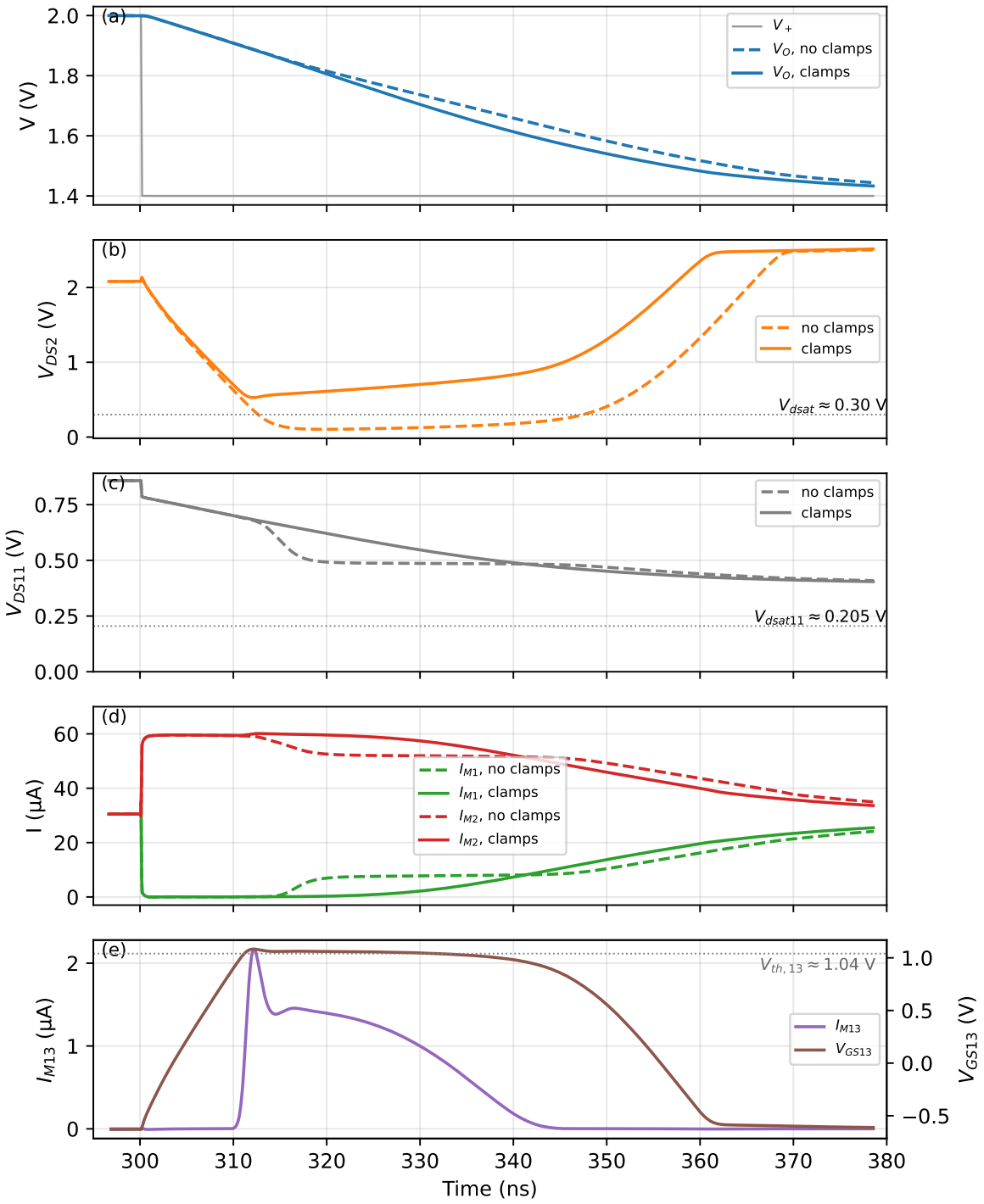


Figure 10: Falling edge of the unity-gain step response with $C_L = 5\,\text{pF}$ (2.0 V to 1.4 V), without clamps (dashed) and with clamps (solid). (a) Input and output. (b) V_{DS2}: without clamps it falls to 104 mV, below V_{dsat}; with clamps it falls to only 0.528 V. (c) V_{DS11}: the tail source stays in saturation in both runs, settling toward 0.40 V as the input returns to 1.4 V; without clamps it sags to about 0.49 V when M_1 turns back on. (d) Input-pair currents: without clamps M_1 conducts about 7 μA during slewing; with clamps it remains off. (e) Clamp M_{13} turns on at about 311 ns, conducting up to 2.2 μA.

Rising Edge From the operating point at $V_+ = 2.0\,\text{V}$, $V_{dsat1} \approx 0.30\,\text{V}$ when M_1 conducts the full tail current. From Fig. 9 we see that without the clamps, V_{DS1} falls to 96 mV, below V_{dsat1};

thus, M_1 enters the triode region. With the clamps, V_{DS1} does not fall below 0.437 V, and M_1 remains in saturation for the entire rising edge. Without the clamps, M_2 turns back on during slewing and conducts about 9 μA, so I_{M1} drops to about 52 μA. With the clamps, M_2 remains off and I_{M1} stays near the full tail current of 60 μA until the input pair begins to rebalance. As a result, the output rises faster, and the rising slew rate increases from 7.21 V/μs to 8.24 V/μs. The clamp M_{12} turns on at about 110 ns, peaks at 2.3 μA, and then conducts about 1.5 μA, tapering to zero by about 142 ns as the input pair rebalances.

Falling Edge Figure 10 shows that without the clamps, V_{DS2} falls to 104 mV, below V_{dsat2}; thus, M_2 enters the triode region. With the clamps, V_{DS2} does not fall below 0.528 V, and M_2 remains in saturation for the entire falling edge. Without the clamps, M_1 conducts about 7 μA during slewing; with the clamps it remains off, and I_{M2} stays near 60 μA until the input pair begins to rebalance. The output falls faster, and the falling slew rate increases from 7.34 V/μs to 8.06 V/μs. The clamp M_{13} turns on at about 311 ns, peaks at 2.2 μA, and tapers to zero by about 341 ns.

In the unity-gain configuration, M_{11} remains in saturation in both cases: without the clamps, the second input transistor turns back on and limits the fall of the tail node; with the clamps, the clamp current keeps M_1 and M_{11} out of the triode region.

8 Final Design Performance

Figure 11 shows the schematic for the final design with the clamps.

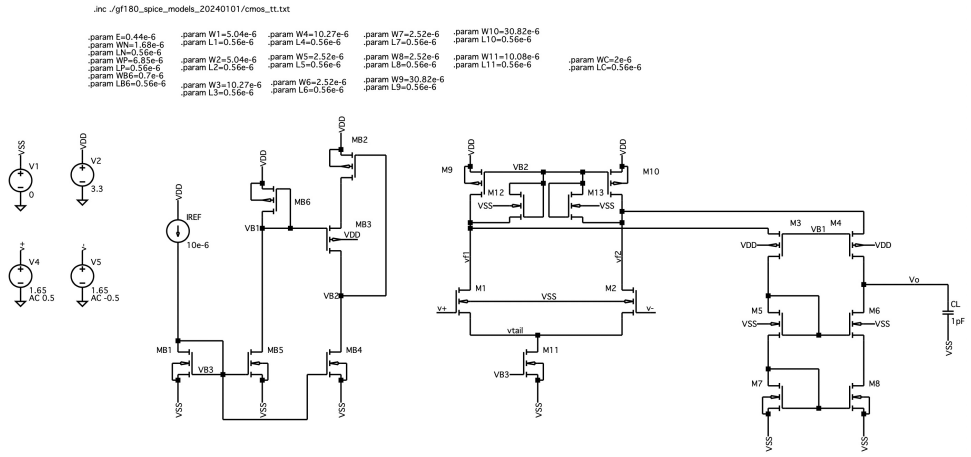


Figure 11: Full Folded-Cascode Schematic with NMOS Clamps

8.1 Frequency Response

Performing the AC analysis using the inductor and capacitor to set the DC operating point again, we have the following frequency response in Fig. 12. The complete netlist is given in Section B.1.

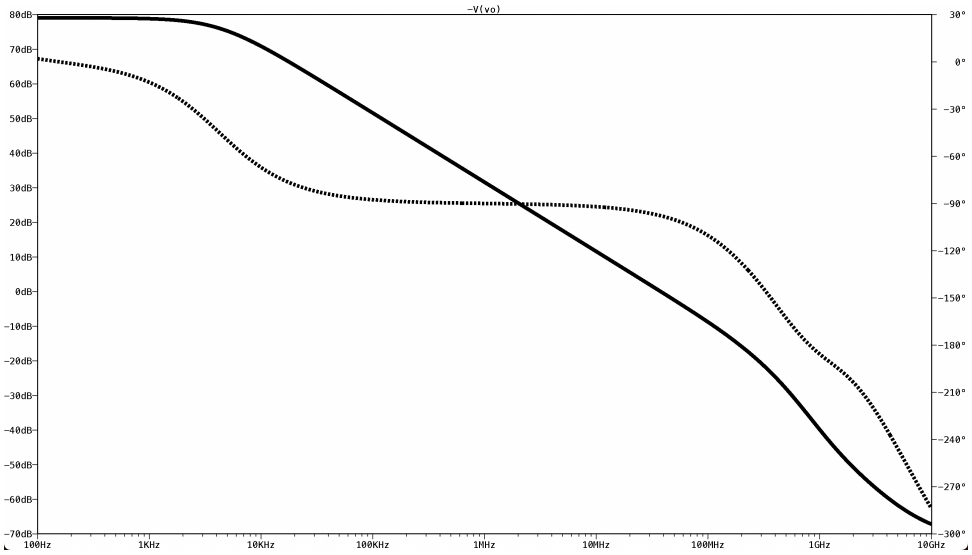


Figure 12: Open-loop Frequency Response with Clamps

Table 8 compares the frequency response with and without the clamps.

Table 8: Open-Loop AC Response With and Without Clamps (TT, 27 °C, $V_{\mathrm{O}} = 1.65\,\mathrm{V}$)

	Without Clamps	With Clamps
DC Gain	79.07 dB	79.07 dB
-3 dB Frequency	4.24 kHz	4.24 kHz
f_{t}	37.87 MHz	37.86 MHz
Phase Margin	82.32°	82.20°
$-180°$ Frequency	835.7 MHz	813.7 MHz
Gain Margin	36.52 dB	36.24 dB

The clamps have no effect on the low-frequency response: the DC gain, $-3\,\mathrm{dB}$ frequency, and unity-gain frequency are unchanged. The capacitance they add at the folding nodes lowers the non-dominant poles slightly, reducing the phase margin by 0.12° and the gain margin by $0.28\,\mathrm{dB}$. The clamps ensure that the input transistors do not go into the triode region during slewing.

8.2 Closed-Loop Settling at $\beta = 0.5$

The settling requirement is based on $\beta = 0.5$, but the step responses of Section 7 used unity gain ($\beta = 1$). We now verify the requirement of $\beta = 0.5$. The op amp is connected as an inverting amplifier with capacitive feedback, $C_{\mathrm{in}} = C_{\mathrm{f}} = 0.5\,\mathrm{pF}$. Therefore, $\beta = C_{\mathrm{f}}/(C_{\mathrm{f}} + C_{\mathrm{in}}) = 0.5$ and the closed-loop gain is $-C_{\mathrm{in}}/C_{\mathrm{f}} = -1\,\frac{\mathrm{V}}{\mathrm{V}}$. A $1\,\mathrm{V}$ input step is used to drive the output from $1.65\,\mathrm{V}$ to $2.65\,\mathrm{V}$ and back. The feedback capacitors add $C_{eq} = \frac{C_f C_{in}}{C_f + C_{in}} = 0.25\,\mathrm{pF}$ of load, for a total of $C_L + C_{eq} = 1.25\,\mathrm{pF}$. Settling is measured against the ideal output ($2.65\,\mathrm{V}$ and $1.65\,\mathrm{V}$). The netlist is given in Section B.3. Figure 13 shows the response:

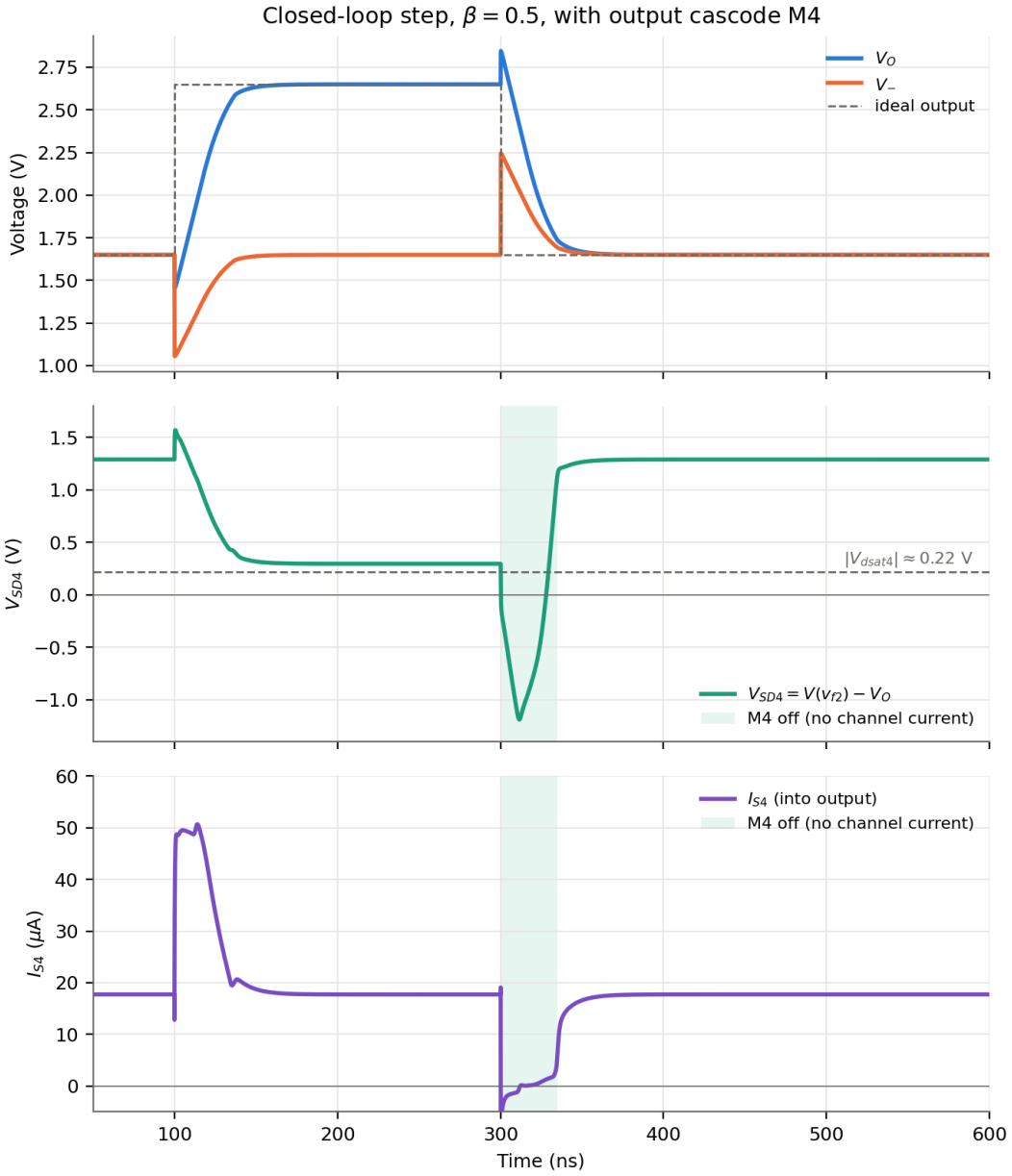


Figure 13: Closed-loop step response at $\beta = 0.5$ ($C_{\text{in}} = C_{\text{f}} = 0.5\,\text{pF}$, $C_L = 1\,\text{pF}$, with clamps). Top: output, node V_- and ideal output. Bottom: error relative to the ideal output; the dotted lines mark where it last enters the 2^{-9} band.

Table 9: Closed-Loop Step Response at $\beta = 0.5$, With and Without Clamps (TT, 27 °C, 1 V Output Step, $C_{\text{in}} = C_{\text{f}} = 0.5\,\text{pF}$, $C_L = 1\,\text{pF}$; peak slope from the simulated waveform)

	Without Clamps	With Clamps
Settling Time to 2^{-9}, Rising (ns)	73.0	73.5
Settling Time to 2^{-9}, Falling (ns)	71.6	70.5
10% to 90% Transition Time, Rising (ns)	27.4	26.4
10% to 90% Transition Time, Falling (ns)	27.8	26.0
10% to 90% Transition Rate, Rising (V/μs)	29.2	30.4
10% to 90% Transition Rate, Falling (V/μs)	28.8	30.8
Peak Slope, Rising (V/μs)	38.7	39.9
Peak Slope, Falling (V/μs)	40.4	40.5
Static Error, Rising (mV)	−0.66	−0.66
Static Error, Falling (mV)	−0.35	−0.35

With the clamps, the amplifier settles to within 2^{-9} of the ideal final value in 73.5 ns on the rising edge and 70.5 ns on the falling edge (Fig. 13 and Table 9), therefore meeting the 100 ns requirement with about 27% margin. This is around twice the unity-gain settling time of Table 7, since halving β halves the closed-loop bandwidth. At each step, the output first moves 0.2 V in the opposite direction, because the input step couples through C_{in} and C_{f} into C_L before the amplifier can respond. The amplifier therefore slews over 1.2 V, with a peak slope of about 40 V/μs. On the falling edge, this jump in the opposite direction moves the output to 2.85 V, just above the upper swing limit of 2.73 V, turning M_4 off. The response shows no overshoot, since phase margin is higher at $\beta = 0.5$. The 0.66 mV static error in Table 9 is the step error of the design and is within the 1.95 mV (2^{-9}) requirement. Random mismatch adds a separate, chip-specific DC shift, which is the input-referred offset multiplied by $1/\beta$. Using the Monte Carlo offset (Section 9.2), this is about ± 44 mV (3σ) at the output. Removing the clamps changes the settling time by at most 1.1 ns and reduces the 10–90% transition rate by only 4–7%, so the design meets the settling requirement with or without them. Most of the settling time is spent in the linear phase, which is not affected by the clamps. The benefit of the clamps is larger with heavier loads, as shown in Section 7.

8.3 CMRR

Figure 14 shows the differential gain, the common-mode gain and the resulting CMRR. The low-frequency CMRR is 73.17 dB and the −3 dB point is near 15.8 MHz, below $f_{\text{t}} = 37.86$ MHz. CMRR begins to roll off at around 10 MHz. This simulation assumes perfectly matched devices, so device mismatch would reduce the CMRR.

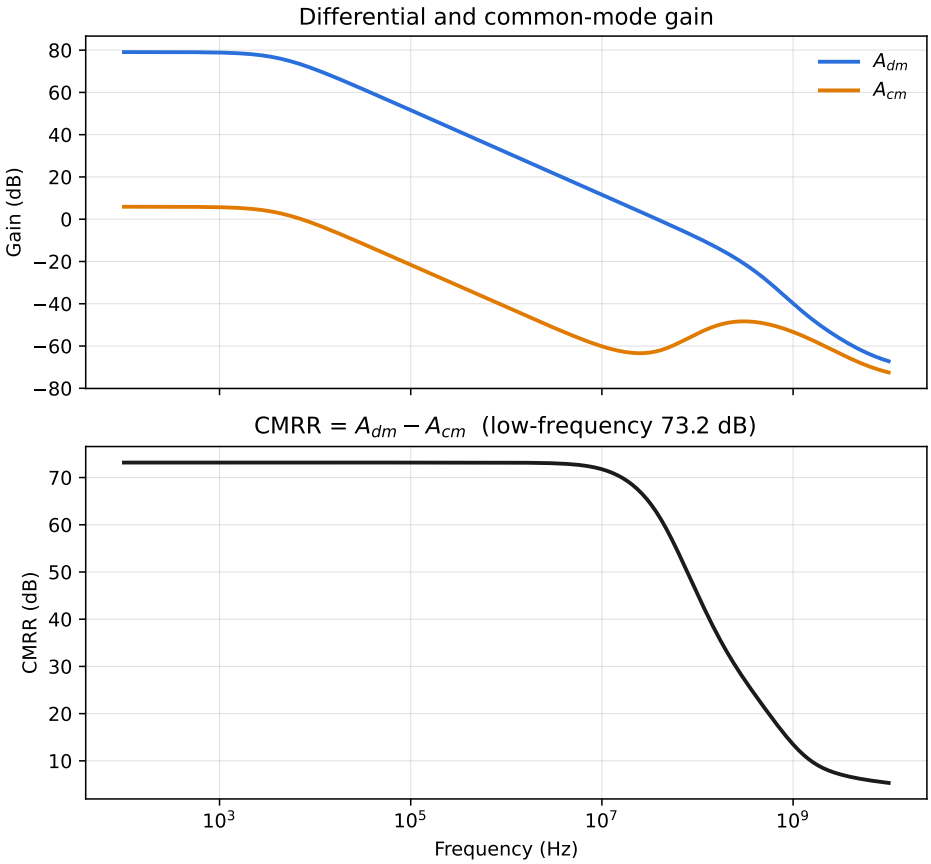


Figure 14: Differential gain A_{dm}, common-mode gain A_{cm} (top) and CMRR $= A_{dm} - A_{cm}$ (bottom), TT, 27 °C.

8.4 Power

Table 10 shows the power consumption summary for the op amp.

Table 10: Power Consumption, TT, 27 °C, $V_{DD} = 3.3$ V

Block	Devices	Current (μA)	Power (μW)	Share
Op amp	M_9, M_{10}	95.5	315.1	75.4%
Bias reference	I_{ref}, M_{B1}	10.0	33.0	7.9%
Bias branch	M_{B2}–M_{B4}	10.6	35.1	8.4%
Bias branch	M_{B5}, M_{B6}	10.4	34.4	8.2%
Total		**126.6**	**417.7**	100%

The total power is 0.418 mW from a 3.3 V supply. The op amp core draws 95.5 μA (75.4%) through M_9 and M_{10}, which carry both the input-pair current and the cascode branches. The bias circuit draws the remaining 31.0 μA in three branches of about 10 μA each.

9 Robustness over Process and Temperature

9.1 Corners

The corner and temperature results were simulated in LTspice with the GF180MCU corner model files, using the inductor/capacitor bench of Section 6 with the clamps and $C_L = 1\,\text{pF}$. The set-up is the AC bench of Section B.1, with the TT model file replaced by the corresponding corner file and the temperature stepped with `.step temp`; the full results are listed in Section A.

We vary the process corner (TT, SS, FF, SF, FS) and the temperature, as shown in Table 11.

Table 11: Variables Swept in the Corner Analysis

Variable	Values
Process Corner	TT, SS, FF, SF, FS
Temperature	−40, 27, 85, 125 °C

Table 12 shows the AC performance by corner, across all temperatures.

Table 12: AC Performance by Corner, Across All Temperatures

Corner	DC Gain (dB)	f_t (MHz)	Phase Margin (°)	Gain Margin (dB)
TT	77.0 to 80.8	30.1 to 46.6	81.6 to 82.7	35.0 to 37.6
SS	76.2 to 80.2	28.5 to 43.9	81.6 to 82.6	35.4 to 38.0
FF	77.1 to 80.9	31.7 to 49.3	81.7 to 82.9	34.7 to 37.4
SF	76.1 to 79.8	28.5 to 43.9	82.1 to 83.0	36.4 to 39.1
FS	77.2 to 81.2	31.6 to 49.3	81.1 to 82.4	33.7 to 36.3
Overall	76.1 to 81.2	28.5 to 49.3	81.1 to 83.0	33.7 to 39.1

Table 13 shows the AC performance by temperature, across all corners.

Table 13: AC Performance by Temperature, Across All Corners

Temperature	DC Gain (dB)	f_t (MHz)	Phase Margin (°)	Gain Margin (dB)
−40 °C	79.8 to 81.2	43.9 to 49.3	81.1 to 82.1	33.7 to 36.4
27 °C	78.2 to 79.4	35.8 to 39.9	81.8 to 82.6	35.0 to 37.7
85 °C	76.9 to 78.1	31.0 to 34.5	82.2 to 82.9	35.8 to 38.6
125 °C	76.1 to 77.2	28.5 to 31.7	82.4 to 83.0	36.3 to 39.1

Table 14 compares the worst-case performance with the specifications.

Table 14: Worst-Case AC Performance Against Specifications

Parameter	Specification	Worst Case	Corner, Temperature
DC Gain	≥ 60.2 dB	76.1 dB	SF, 125 °C
f_t	≥ 19.86 MHz	28.5 MHz	SS, SF, 125 °C
Phase Margin	–	81.1°	FS, −40 °C
Gain Margin	–	33.7 dB	FS, −40 °C

Figure 15 shows the DC gain, unity-gain frequency, phase margin and gain margin across all corners and temperatures.

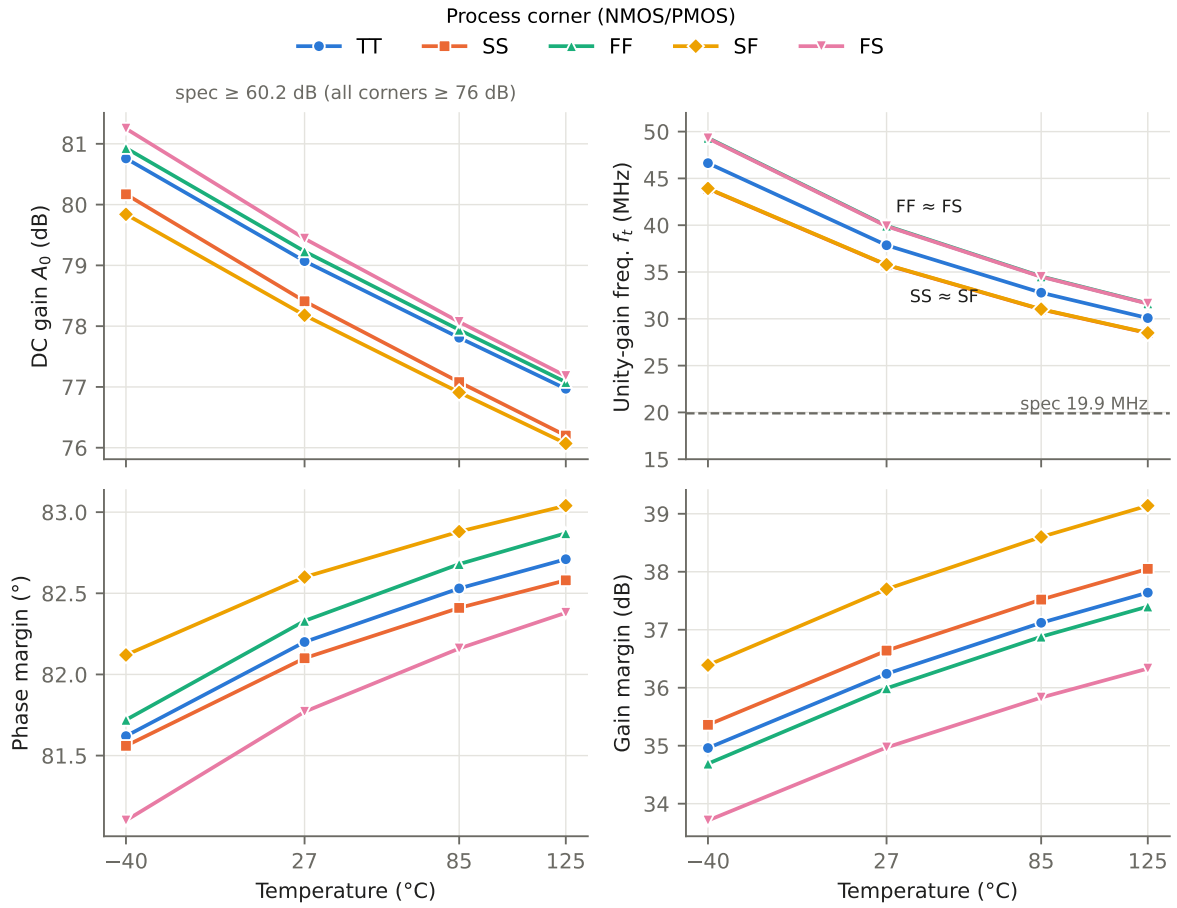


Figure 15: Folded-cascode AC performance across process corners and temperature. f_t is set by the NMOS corner (SS≈SF, FF≈FS). Worst case: $A_0 = 76.1$ dB (SF, 125 °C), $f_t = 28.5$ MHz (SS, SF, 125 °C), PM = 81.1° and GM = 33.7 dB (FS, −40 °C).

The design meets both specifications at every corner and temperature, with at least 15.9 dB of margin on the DC gain and 43% on the unity-gain frequency. Temperature has the largest effect: as mobility falls, f_t decreases by about 36% from −40 °C to 125 °C and the DC gain by about 4 dB. Because the folded-cascode is a single-stage amplifier compensated by its load, its phase margin varies by less than 2° across all corners.

9.2 Monte Carlo

The corner analysis of Section 9.1 moves every NMOS device together and every PMOS device together. As a result, it does not show random mismatch between identical devices. Monte Carlo analysis is used to capture this effect. The analysis is run in ngspice-47 with the full PDK library (`sm141064.ngspice`), using the same inductor/capacitor bench as Section 6. The PDK provides two independent switches [2]:

- `sw_stat_global`: process variation shared by all devices of the same type (all NMOS move together, all PMOS move together), as in a corner analysis but drawn from a distribution.
- `sw_stat_mismatch`: independent variation of each device. The threshold-voltage mismatch follows Pelgrom's rule, $\sigma_{V_T} = 0.7071\, A_{VT}/\sqrt{L_{\text{eff}} W_{\text{eff}}}$, with $A_{VT} = 7.148$ mV µm (NMOS) and 6.66 mV µm (PMOS), $L_{\text{eff}} = L - 0.15$ µm and $W_{\text{eff}} = W + 0.1$ µm.

Two Monte Carlo runs are performed, each with 500 trials at TT and 27 °C and a fixed

random seed (`setseed 1`) so that they can be repeated: one with mismatch only, and one with global plus mismatch, which is the realistic case.

The netlist is given in Section B.4. Table 15 summarizes both runs, and Figs. 16 and 17 show the histograms.

Table 15: Monte Carlo Results, 500 Trials, TT, 27 °C (mean μ, standard deviation σ, and range)

	Mismatch Only		Global + Mismatch	
	$\mu \pm \sigma$	Range	$\mu \pm \sigma$	Range
V_{OS} (mV)	0.89 ± 7.26	-19.6 to 27.0	0.85 ± 7.31	-20.3 to 28.3
DC Gain (dB)	79.09 ± 1.68	74.10 to 84.13	79.14 ± 3.46	67.31 to 90.71
f_t (MHz)	37.76 ± 1.16	34.39 to 41.31	37.54 ± 2.08	21.89 to 43.73
Phase Margin (°)	82.22 ± 0.64	79.75 to 83.65	81.76 ± 2.96	37.30 to 84.95
Gain Margin (dB)	35.62 ± 0.74	32.86 to 37.58	35.36 ± 2.18	15.74 to 39.60
V_{SD9} (V)	0.358 ± 0.023	0.289 to 0.431	0.360 ± 0.032	0.261 to 0.535
Power (mW)	0.418 ± 0.026	0.344 to 0.489	0.417 ± 0.041	0.283 to 0.539

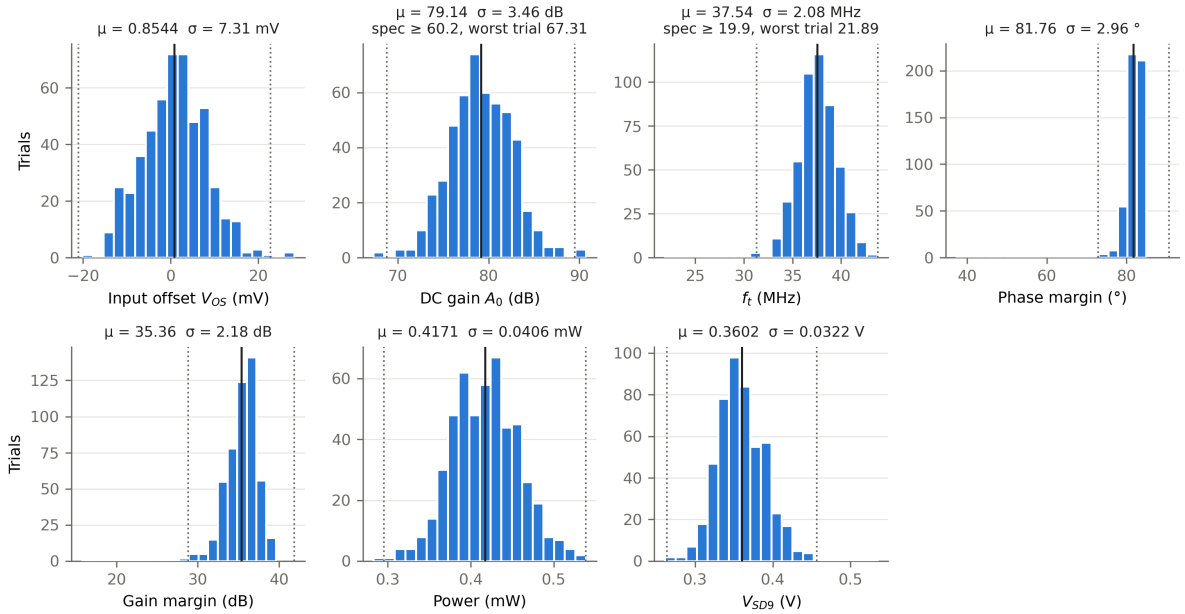


Figure 16: Monte Carlo histograms, global plus mismatch variation, 500 trials, TT, 27 °C. The solid line marks the mean μ and the dotted lines $\mu \pm 3\sigma$.

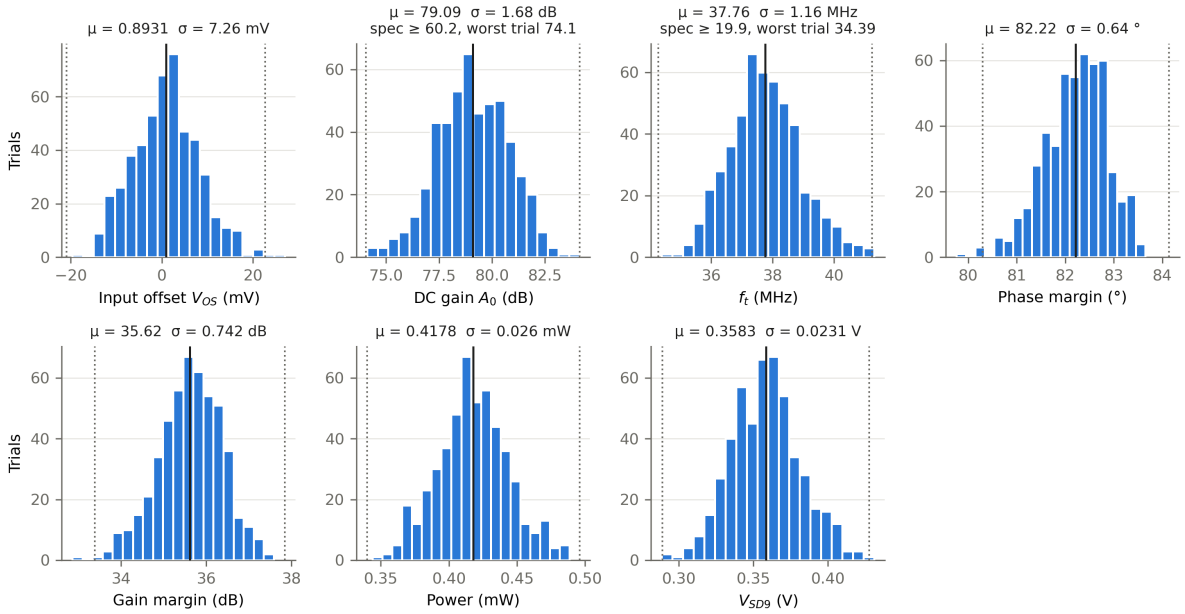


Figure 17: Monte Carlo histograms, mismatch only, 500 trials, TT, 27 °C. The solid line marks the mean μ and the dotted lines $\mu \pm 3\sigma$.

Input Offset The input offset has a standard deviation of 7.26 mV with mismatch only and 7.31 mV with global variation added. Since the two values are similar, the offset comes mainly from mismatch between devices. In both runs the mean offset is close to zero, so the $\pm 3\sigma$ range for the global-plus-mismatch run is about ± 22 mV.

Gain and Bandwidth The DC gain varies more when global variation is included as its standard deviation increases from 1.68 dB to 3.46 dB. The lowest gain in the 500 trials is 67.31 dB, slightly below the $\mu - 3\sigma$ value of 68.8 dB, and well above the 60.2 dB specification. The unity-gain frequency has a standard deviation of 2.1 MHz and a $\mu - 3\sigma$ value of 31.3 MHz; one trial falls well below this, at 21.9 MHz, which still meets the 19.86 MHz specification. No trial fails either specification.

Phase and Gain Margin The phase margin has a mean of 81.8° and a standard deviation of 2.96°. Unlike the offset and DC gain, its distribution has a long tail on the low side. Seven trials fall below 75°, and the worst has a phase margin of 37.3°, a gain margin of 15.7 dB and $f_t = 21.9$ MHz. In this trial, the cascode branch is almost starved of current. With very little current, $g_{m3,4}$ is very small, so the pole at the folding nodes moves down close to f_t, reducing the phase margin and f_t, while the higher output resistance raises the DC gain to 90.7 dB.

The global variation raised the tail current to 67.3 μA and lowered the folding current I_B to 34.0 μA, leaving $I_B - I/2 \approx 0.3\,\mu$A in $M_{3,4}$ instead of about 18 μA. Both mirrors that set these currents ($M_{B1} \to M_{11}$ and $M_{B2} \to M_{9,10}$) pair a narrow bias device with a wide output device in different model bins, which increases the global spread of I and I_B. All trials still meet the gain and unity-gain frequency specifications, but the margin $I_B - I/2 = 15\,\mu$A sets a limit on robustness; a larger ratio $I_B/(I/2)$ or geometry-matched bias mirrors would remove this tail.

10 Conclusion

A folded-cascode op amp was designed in the GF180MCU process using a V_{OV}-based sizing method. At TT and 27 °C the design achieves a DC gain of 79.07 dB and a unity-gain frequency

of 37.9 MHz, both exceeding the 60.2 dB and 19.86 MHz requirements, along with a phase margin of 82.2°, a gain margin of 36.2 dB and a power consumption of 0.418 mW.

In the closed-loop configuration with $\beta = 0.5$, the amplifier settles to within 2^{-9} of a 1 V step in 73.5 ns (rising) and 70.5 ns (falling), within the 100 ns requirement. The NMOS clamps change the settling times by at most 1.1 ns because most of the settling time is spent in the linear phase, which is not affected by the clamps. The narrow-diode bias circuit design keeps M_9 and M_{10} 150 mV above saturation, leading to an improvement in the output swing compared to the stacked-diode bias circuit. The NMOS clamps help keep M_1, M_2 and M_{11} out of the triode region during slewing. They draw no static current and reduce the phase margin by only 0.12°. In a unity-gain feedback configuration, the use of the NMOS clamps increased the rising slew rate from 39.3 to 44.4 V/μs with a 1 pF load, and from 7.2 to 8.2 V/μs with 5 pF.

Across all process corners and temperatures from −40 to 125 °C, the design meets both specifications, with a worst-case DC gain of 76.1 dB and unity-gain frequency of 28.5 MHz. Temperature causes a larger variation than the process corners: from −40 to 125 °C, f_t falls by 36%, compared with a spread of about 11% across the five corners at 27 °C.

In the Monte Carlo simulations no trial failed either specification: the lowest DC gain was 67.31 dB and the lowest unity-gain frequency 21.89 MHz. The mean DC gain of 79.14 dB is close to the simulated 79.07 dB. Global variation increased the spread of the DC gain and phase margin, while the input offset was set mainly by device mismatch, with a mean close to zero and a standard deviation of 7.31 mV. The phase margin has a long low tail, reaching 37.3° in one trial where global variation left almost no current in the cascode branch; increasing I_{B} relative to $I/2$, or matching the geometry of the bias mirrors, would remove this tail.

The main limitation of the design is the random input offset, which would produce a DC output offset of about ±44 mV (3σ) at the target feedback factor of $\beta = 0.5$.

References

[1] A. S. Sedra and K. C. Smith, *Microelectronic Circuits*, 7th ed. Oxford University Press, 2015.

[2] GlobalFoundries, “GF180MCU open-source PDK,” model release 2024-01-01.

A Full Corner Data

Table 16 lists the AC results for every process corner and temperature, from the LTspice inductor/capacitor bench with $C_L = 1$ pF.

Table 16: Corner Data, Folded-Cascode Op Amp

Corner	T (°C)	A_0 (dB)	f_{3dB} (kHz)	f_t (MHz)	PM (°)	f_{180} (MHz)	GM (dB)
TT	−40	80.8	4.31	46.6	81.6	900	35.0
	27	79.1	4.24	37.9	82.2	814	36.2
	85	77.8	4.25	32.8	82.5	757	37.1
	125	77.0	4.29	30.1	82.7	724	37.6
SS	−40	80.2	4.34	43.9	81.6	864	35.4
	27	78.4	4.33	35.8	82.1	783	36.6
	85	77.1	4.37	31.0	82.4	728	37.5
	125	76.2	4.44	28.5	82.6	698	38.0
FF	−40	80.9	4.47	49.3	81.7	942	34.7
	27	79.2	4.40	39.9	82.3	852	36.0
	85	77.9	4.41	34.5	82.7	793	36.9
	125	77.1	4.46	31.7	82.9	759	37.4
SF	−40	79.8	4.51	43.9	82.1	954	36.4
	27	78.2	4.44	35.8	82.6	866	37.7
	85	76.9	4.46	31.0	82.9	807	38.6
	125	76.1	4.51	28.5	83.0	774	39.1
FS	−40	81.2	4.31	49.3	81.1	856	33.7
	27	79.4	4.29	39.9	81.8	774	35.0
	85	78.1	4.34	34.5	82.2	718	35.8
	125	77.2	4.41	31.6	82.4	687	36.3

B SPICE Netlists

B.1 Folded-Cascode Op Amp and AC Test Bench

The netlist below was simulated in LTspice with the GF180MCU models at TT and 27 °C.

```
* Folded-cascode op amp, GF180MCU 3.3 V, AC test bench (TT, 27 C)
* LFB/CFB close the loop at DC (Vout = 1.65 V) and open it at AC,
* so V(vo)/V(vs) = -A.

* Supplies
V1 VSS 0 0
V2 VDD 0 3.3

* op amp
M11 vtail VB3 VSS VSS nmos_3p3 l={L11} w={W11} ad={W11*E} as={W11*E} pd={2*W11+2*E} ps={2*W11+2*E} m=1
M1 vf1 v+ vtail VSS nmos_3p3 l={L1} w={W1} ad={W1*E} as={W1*E} pd={2*W1+2*E} ps={2*W1+2*E} m=1
M2 vf2 v- vtail VSS nmos_3p3 l={L2} w={W2} ad={W2*E} as={W2*E} pd={2*W2+2*E} ps={2*W2+2*E} m=1
M9 vf1 VB2 VDD VDD pmos_3p3 l={L9} w={W9} ad={W9*E} as={W9*E} pd={2*W9+2*E} ps={2*W9+2*E} m=1
M10 vf2 VB2 VDD VDD pmos_3p3 l={L10} w={W10} ad={W10*E} as={W10*E} pd={2*W10+2*E} ps={2*W10+2*E} m=1
M3 N002 VB1 vf1 VDD pmos_3p3 l={L3} w={W3} ad={W3*E} as={W3*E} pd={2*W3+2*E} ps={2*W3+2*E} m=1
M4 Vo VB1 vf2 VDD pmos_3p3 l={L4} w={W4} ad={W4*E} as={W4*E} pd={2*W4+2*E} ps={2*W4+2*E} m=1
M5 N002 N002 N004 VSS nmos_3p3 l={L5} w={W5} ad={W5*E} as={W5*E} pd={2*W5+2*E} ps={2*W5+2*E} m=1
M6 Vo N002 N005 VSS nmos_3p3 l={L6} w={W6} ad={W6*E} as={W6*E} pd={2*W6+2*E} ps={2*W6+2*E} m=1
M7 N004 N004 VSS VSS nmos_3p3 l={L7} w={W7} ad={W7*E} as={W7*E} pd={2*W7+2*E} ps={2*W7+2*E} m=1
M8 N005 N004 VSS VSS nmos_3p3 l={L8} w={W8} ad={W8*E} as={W8*E} pd={2*W8+2*E} ps={2*W8+2*E} m=1
CL Vo VSS 1pF

* Clamps
M12 VB2 VB2 vf1 VSS nmos_3p3 l={LC} w={WC} ad={WC*E} as={WC*E} pd={2*WC+2*E} ps={2*WC+2*E} m=1
M13 VB2 VB2 vf2 VSS nmos_3p3 l={LC} w={WC} ad={WC*E} as={WC*E} pd={2*WC+2*E} ps={2*WC+2*E} m=1

* Bias circuit
IREF VDD VB3 10e-6
MB1 VB3 VB3 VSS VSS nmos_3p3 l={LN} w={WN} ad={WN*E} as={WN*E} pd={2*WN+2*E} ps={2*WN+2*E} m=1
MB2 N001 VB2 VDD VDD pmos_3p3 l={LP} w={WP} ad={WP*E} as={WP*E} pd={2*WP+2*E} ps={2*WP+2*E} m=1
MB3 VB2 VB1 N001 VDD pmos_3p3 l={LP} w={WP} ad={WP*E} as={WP*E} pd={2*WP+2*E} ps={2*WP+2*E} m=1
MB4 VB2 VB3 VSS VSS nmos_3p3 l={LN} w={WN} ad={WN*E} as={WN*E} pd={2*WN+2*E} ps={2*WN+2*E} m=1
MB5 VB1 VB3 VSS VSS nmos_3p3 l={LN} w={WN} ad={WN*E} as={WN*E} pd={2*WN+2*E} ps={2*WN+2*E} m=1
MB6 VB1 VB1 VDD VDD pmos_3p3 l={LB6} w={WB6} ad={WB6*E} as={WB6*E} pd={2*WB6+2*E} ps={2*WB6+2*E} m=1
```

```
* Test bench
vs N003 0 0 AC 1
CFB v- N003 1e6
LFB Vo v- 1e6
V3 v+ 0 1.65

* Models and sizes
.inc ./gf180_spice_models_20240101/cmos_tt.txt
.param E=0.44e-6
.param WN=1.68e-6 LN=0.56e-6
.param WP=6.85e-6 LP=0.56e-6
.param WB6=0.7e-6 LB6=0.56e-6
.param W1=5.04e-6 L1=0.56e-6
.param W2=5.04e-6 L2=0.56e-6
.param W3=10.27e-6 L3=0.56e-6
.param W4=10.27e-6 L4=0.56e-6
.param W5=2.52e-6 L5=0.56e-6
.param W6=2.52e-6 L6=0.56e-6
.param W7=2.52e-6 L7=0.56e-6
.param W8=2.52e-6 L8=0.56e-6
.param W9=30.82e-6 L9=0.56e-6
.param W10=30.82e-6 L10=0.56e-6
.param W11=10.08e-6 L11=0.56e-6
.param WC=2e-6 LC=0.56e-6

* Analysis and measurements
.ac dec 100 100 10G
.options meascplxfmt=cartesian
.meas AC a0 MAX mag(V(vo))
.meas AC a0db PARAM 20*log10(a0)
.meas AC f3db WHEN mag(V(vo))=a0/sqrt(2) FALL=1
.meas AC ft WHEN mag(V(vo))=1
.meas AC ph FIND ph(-V(vo)) WHEN mag(V(vo))=1
.meas AC pm PARAM 180+ph
.meas AC f180 WHEN Im(V(vo))=0 FALL=1
.meas AC g180 FIND mag(V(vo)) WHEN Im(V(vo))=0 FALL=1
.meas AC gmdb PARAM -20*log10(g180)
.end
```

B.2 Unity-Gain Step Response Test Bench

The unity-gain step responses in Section 7 use the netlist of Section B.1 with the gate of M_2 (the inverting input) tied to the output and the non-inverting input driven through 1 kΩ. The listing below is the run without clamps; the run with clamps keeps M_{12} and M_{13} as in Section B.1. The 1 pF load uses a 1 V step from 1.5 to 2.5 V; the 5 pF load uses a 0.6 V step from 1.4 to 2.0 V, with the measurement levels scaled to match. Settling is measured against the input V_+. The lines that differ are:

```
* M2 gate tied to the output (unity-gain buffer)
M2 vf2 Vo vtail VSS nmos_3p3 l={L2} w={W2} ad={W2*E} as={W2*E} pd={2*W2+2*E} ps={2*W2+2*E} m=1
* Delete: vs, CFB, LFB, V3 and the .ac measurements (and M12, M13 for the no-clamp run)
* Input step through 1 kohm into the M1 gate (v+)
V5 N003 0 PULSE(1.5 2.5 0.1u 100p 100p 200n)
* 5 pF case: V5 N003 0 PULSE(1.4 2.0 0.1u 100p 100p 200n) and CL Vo VSS 5pF
R1 v+ N003 1000

* overshoot / undershoot
.meas tran vlo AVG V(vo) FROM 80n TO 100n
.meas tran vhi AVG V(vo) FROM 280n TO 300n
.meas tran vpk MAX V(vo) FROM 100n TO 300n
.meas tran vmin MIN V(vo) FROM 300n TO 500n
.meas tran os_mV PARAM (vpk-vhi)*1e3
.meas tran os_pct PARAM 100*(vpk-vhi)/(vhi-vlo)
.meas tran us_mV PARAM (vlo-vmin)*1e3
.meas tran us_pct PARAM 100*(vlo-vmin)/(vhi-vlo)
* 10-90% edges and slew rate (V/us), 1 V step
.meas tran trise TRIG V(vo)=1.6 RISE=1 TARG V(vo)=2.4 RISE=1
.meas tran tfall TRIG V(vo)=2.4 FALL=1 TARG V(vo)=1.6 FALL=1
.meas tran sr_rise PARAM 0.8e-6/trise
```

```
.meas tran sr_fall PARAM 0.8e-6/tfall
* folding nodes during slewing
.meas tran f1min MIN V(vf1) FROM 100n TO 200n
.meas tran f2min MIN V(vf2) FROM 300n TO 400n
* M1 drain-source voltage, tail node, and M2 current on the rising edge
.meas tran vds1min MIN V(vf1,vtail) FROM 100n TO 200n
.meas tran tailmin MIN V(vtail) FROM 102n TO 200n
.meas tran id2mid FIND Id(M2) WHEN V(vo)=2.0 RISE=1
* settling error against the input for each edge (zero outside its window)
Ber er 0 V=abs(V(vo)-V(v+))*u(300n-time)*u(time-100n)
Bef ef 0 V=abs(V(vo)-V(v+))*u(time-300.1n)
* settling to 2^-9 of a 1 V step = 1.953 mV
.meas tran tsr_abs WHEN V(er)=1.953m CROSS=LAST
.meas tran tsf_abs WHEN V(ef)=1.953m CROSS=LAST
.meas tran tset_rise_ns PARAM (tsr_abs-100n)*1e9
.meas tran tset_fall_ns PARAM (tsf_abs-300.1n)*1e9

.tran 0 600n 0 50p
```

B.3 Closed-Loop Step Response Test Bench ($\beta = 0.5$)

The settling time and transition rate at the target feedback factor are simulated with the op amp connected as an inverting amplifier with capacitive feedback ($C_{\text{in}} = C_{\text{f}} = 0.5\,\text{pF}$, $\beta = 0.5$, closed-loop gain -1), driven by a 1 V input step. The listing below is the final design with clamps; the run without clamps removes M_{12} and M_{13}.

```
* Folded-cascode op amp, GF180MCU 3.3 V, closed-loop step response (TT, 27 C)
* Inverting amplifier, CIN = CF = 0.5 pF: beta = CF/(CF+CIN) = 0.5, gain = -1.
* 1 V input step down at 100 ns and back up at 300 ns, so the output steps
* 1.65 V -> 2.65 V -> 1.65 V. RF = 1 TOhm sets the DC operating point only.

* Supplies
V1 VSS 0 0
V2 VDD 0 3.3

* op amp (M2 gate = inverting input v-)
M11 vtail VB3 VSS VSS nmos_3p3 l={L11} w={W11} ad={W11*E} as={W11*E} pd={2*W11+2*E} ps={2*W11+2*E} m=1
M1 vf1 v+ vtail VSS nmos_3p3 l={L1} w={W1} ad={W1*E} as={W1*E} pd={2*W1+2*E} ps={2*W1+2*E} m=1
M2 vf2 v- vtail VSS nmos_3p3 l={L2} w={W2} ad={W2*E} as={W2*E} pd={2*W2+2*E} ps={2*W2+2*E} m=1
M9 vf1 VB2 VDD VDD pmos_3p3 l={L9} w={W9} ad={W9*E} as={W9*E} pd={2*W9+2*E} ps={2*W9+2*E} m=1
M10 vf2 VB2 VDD VDD pmos_3p3 l={L10} w={W10} ad={W10*E} as={W10*E} pd={2*W10+2*E} ps={2*W10+2*E} m=1
M3 N002 VB1 vf1 VDD pmos_3p3 l={L3} w={W3} ad={W3*E} as={W3*E} pd={2*W3+2*E} ps={2*W3+2*E} m=1
M4 Vo VB1 vf2 VDD pmos_3p3 l={L4} w={W4} ad={W4*E} as={W4*E} pd={2*W4+2*E} ps={2*W4+2*E} m=1
M5 N002 N002 N003 VSS nmos_3p3 l={L5} w={W5} ad={W5*E} as={W5*E} pd={2*W5+2*E} ps={2*W5+2*E} m=1
M6 Vo N002 N004 VSS nmos_3p3 l={L6} w={W6} ad={W6*E} as={W6*E} pd={2*W6+2*E} ps={2*W6+2*E} m=1
M7 N003 N003 VSS VSS nmos_3p3 l={L7} w={W7} ad={W7*E} as={W7*E} pd={2*W7+2*E} ps={2*W7+2*E} m=1
M8 N004 N003 VSS VSS nmos_3p3 l={L8} w={W8} ad={W8*E} as={W8*E} pd={2*W8+2*E} ps={2*W8+2*E} m=1
CL Vo VSS 1pF

* Clamps (removed for the no-clamp run)
M12 VB2 VB2 vf1 VSS nmos_3p3 l={LC} w={WC} ad={WC*E} as={WC*E} pd={2*WC+2*E} ps={2*WC+2*E} m=1
M13 VB2 VB2 vf2 VSS nmos_3p3 l={LC} w={WC} ad={WC*E} as={WC*E} pd={2*WC+2*E} ps={2*WC+2*E} m=1

* Bias circuit
IREF VDD VB3 10e-6
MB1 VB3 VB3 VSS VSS nmos_3p3 l={LN} w={WN} ad={WN*E} as={WN*E} pd={2*WN+2*E} ps={2*WN+2*E} m=1
MB2 N001 VB2 VDD VDD pmos_3p3 l={LP} w={WP} ad={WP*E} as={WP*E} pd={2*WP+2*E} ps={2*WP+2*E} m=1
MB3 VB2 VB1 N001 VDD pmos_3p3 l={LP} w={WP} ad={WP*E} as={WP*E} pd={2*WP+2*E} ps={2*WP+2*E} m=1
MB4 VB2 VB3 VSS VSS nmos_3p3 l={LN} w={WN} ad={WN*E} as={WN*E} pd={2*WN+2*E} ps={2*WN+2*E} m=1
MB5 VB1 VB3 VSS VSS nmos_3p3 l={LN} w={WN} ad={WN*E} as={WN*E} pd={2*WN+2*E} ps={2*WN+2*E} m=1
MB6 VB1 VB1 VDD VDD pmos_3p3 l={LB6} w={WB6} ad={WB6*E} as={WB6*E} pd={2*WB6+2*E} ps={2*WB6+2*E} m=1

* Test bench: inverting amplifier with capacitive feedback
V3 N005 0 PULSE(1.65 0.65 0.1u 100p 100p 200n)
V4 v+ 0 1.65
CIN v- N005 0.5p
CF v- Vo 0.5p
RF v- Vo 1e12

* Models and sizes
```

```
.inc ./gf180_spice_models_20240101/cmos_tt.txt
.param E=0.44e-6
.param WN=1.68e-6 LN=0.56e-6
.param WP=6.85e-6 LP=0.56e-6
.param WB6=0.7e-6 LB6=0.56e-6
.param W1=5.04e-6 L1=0.56e-6
.param W2=5.04e-6 L2=0.56e-6
.param W3=10.27e-6 L3=0.56e-6
.param W4=10.27e-6 L4=0.56e-6
.param W5=2.52e-6 L5=0.56e-6
.param W6=2.52e-6 L6=0.56e-6
.param W7=2.52e-6 L7=0.56e-6
.param W8=2.52e-6 L8=0.56e-6
.param W9=30.82e-6 L9=0.56e-6
.param W10=30.82e-6 L10=0.56e-6
.param W11=10.08e-6 L11=0.56e-6
.param WC=2e-6 LC=0.56e-6

* Overshoot / undershoot relative to the ideal final values
.meas tran vpk MAX V(vo) FROM 100n TO 300n
.meas tran vmin MIN V(vo) FROM 300n TO 500n
.meas tran os_mV PARAM (vpk-2.65)*1e3
.meas tran us_mV PARAM (1.65-vmin)*1e3

* 10-90% transition times and rates (V/us), 1 V output step
.meas tran trise TRIG V(vo)=1.75 RISE=1 TARG V(vo)=2.55 RISE=1
.meas tran tfall TRIG V(vo)=2.55 FALL=1 TARG V(vo)=1.75 FALL=1
.meas tran sr_rise PARAM 0.8e-6/trise
.meas tran sr_fall PARAM 0.8e-6/tfall

* Settling error against the ideal final values (2.65 V and 1.65 V)
Ber er 0 V=abs(V(vo)-2.65)*u(300n-time)*u(time-100n)
Bef ef 0 V=abs(V(vo)-1.65)*u(time-300.1n)

* Settling to 2^-9 of the 1 V step = 1.953 mV
.meas tran tsr_abs WHEN V(er)=1.953m CROSS=LAST
.meas tran tsf_abs WHEN V(ef)=1.953m CROSS=LAST
.meas tran tset_rise_ns PARAM (tsr_abs-100n)*1e9
.meas tran tset_fall_ns PARAM (tsf_abs-300.1n)*1e9

* Static error just before each edge ends
.meas tran verr_rise_mV FIND (V(vo)-2.65)*1e3 AT 295n
.meas tran verr_fall_mV FIND (V(vo)-1.65)*1e3 AT 495n

* Analysis
.tran 0 600n 0 50p
.end
```

B.4 Monte Carlo Set-Up (ngspice)

The Monte Carlo runs use ngspice-47 with the full GF180MCU library, which contains the statistical model sections. The PDK defines the transistors as subcircuits, so each device has an `X` prefix. The listing below is the global-plus-mismatch run; the mismatch-only run is identical except for `sw_stat_global=0`.

```
* Folded-cascode op amp: Monte Carlo of offset, headroom, A0, ft, PM, GM
* Statistical switches: 0/1 = mismatch only, 1/0 = global only, 1/1 = realistic
.param sw_stat_global=1 sw_stat_mismatch=1
.param mc_skew=3 res_mc_skew=3 cap_mc_skew=3 fnoicor=0
.lib sm141064.ngspice statistical
.temp 27
.param E=0.44u
.param L0=0.56u
.param W1=5.04u W3=10.27u W5=2.52u W9=30.82u W11=10.08u
.param WN=1.68u WP=6.85u WB6=0.7u WC=2u

V1 VSS 0 0
V2 VDD 0 3.3

* ---- bias network ----
```

```
IREF VDD VB3 10u
XMB1 VB3 VB3 VSS VSS nmos_3p3 w={WN} l={L0} ad={WN*E} as={WN*E} pd={2*WN+2*E} ps={2*WN+2*E}
XMB4 VB2 VB3 VSS VSS nmos_3p3 w={WN} l={L0} ad={WN*E} as={WN*E} pd={2*WN+2*E} ps={2*WN+2*E}
XMB5 VB1 VB3 VSS VSS nmos_3p3 w={WN} l={L0} ad={WN*E} as={WN*E} pd={2*WN+2*E} ps={2*WN+2*E}
XMB2 NB2 VB2 VDD VDD pmos_3p3 w={WP} l={L0} ad={WP*E} as={WP*E} pd={2*WP+2*E} ps={2*WP+2*E}
XMB3 VB2 VB1 NB2 VDD pmos_3p3 w={WP} l={L0} ad={WP*E} as={WP*E} pd={2*WP+2*E} ps={2*WP+2*E}
XMB6 VB1 VB1 VDD VDD pmos_3p3 w={WB6} l={L0} ad={WB6*E} as={WB6*E} pd={2*WB6+2*E} ps={2*WB6+2*E}

* ---- core (M1 = non-inverting, M2 = inverting input) ----
XM11 vtail VB3 VSS VSS nmos_3p3 w={W11} l={L0} ad={W11*E} as={W11*E} pd={2*W11+2*E} ps={2*W11+2*E}
XM1 vf1 vp vtail VSS nmos_3p3 w={W1} l={L0} ad={W1*E} as={W1*E} pd={2*W1+2*E} ps={2*W1+2*E}
XM2 vf2 vm vtail VSS nmos_3p3 w={W1} l={L0} ad={W1*E} as={W1*E} pd={2*W1+2*E} ps={2*W1+2*E}
XM9 vf1 VB2 VDD VDD pmos_3p3 w={W9} l={L0} ad={W9*E} as={W9*E} pd={2*W9+2*E} ps={2*W9+2*E}
XM10 vf2 VB2 VDD VDD pmos_3p3 w={W9} l={L0} ad={W9*E} as={W9*E} pd={2*W9+2*E} ps={2*W9+2*E}
XM3 vmir VB1 vf1 VDD pmos_3p3 w={W3} l={L0} ad={W3*E} as={W3*E} pd={2*W3+2*E} ps={2*W3+2*E}
XM4 Vo VB1 vf2 VDD pmos_3p3 w={W3} l={L0} ad={W3*E} as={W3*E} pd={2*W3+2*E} ps={2*W3+2*E}
XM5 vmir vmir vs5 VSS nmos_3p3 w={W5} l={L0} ad={W5*E} as={W5*E} pd={2*W5+2*E} ps={2*W5+2*E}
XM6 Vo vmir vs6 VSS nmos_3p3 w={W5} l={L0} ad={W5*E} as={W5*E} pd={2*W5+2*E} ps={2*W5+2*E}
XM7 vs5 vs5 VSS VSS nmos_3p3 w={W5} l={L0} ad={W5*E} as={W5*E} pd={2*W5+2*E} ps={2*W5+2*E}
XM8 vs6 vs5 VSS VSS nmos_3p3 w={W5} l={L0} ad={W5*E} as={W5*E} pd={2*W5+2*E} ps={2*W5+2*E}
* clamps
XM12 VB2 VB2 vf1 VSS nmos_3p3 w={WC} l={L0} ad={WC*E} as={WC*E} pd={2*WC+2*E} ps={2*WC+2*E}
XM13 VB2 VB2 vf2 VSS nmos_3p3 w={WC} l={L0} ad={WC*E} as={WC*E} pd={2*WC+2*E} ps={2*WC+2*E}
CL Vo VSS 1p

* ---- bench: DC feedback through L, AC drive into v- through C ----
* DC: C open, L short -> unity-gain buffer, output at ~1.65 V
* AC: L open, C short -> vs drives v- directly, vo/vs = -A (open-loop gain)
Vcm vp 0 dc 1.65
LFB Vo vm 1e6
CFB vm vs 1e6
Vs vs 0 dc 0 ac 1

.control
set units = degrees
setseed 1
echo "vos_V vsd9_V vsd10_V vsd3_V vsd4_V vds1_V vds11_V ptot_W a0_dB ft_Hz pm_deg gm_dB" >
     fc_mc_11_results.txt
repeat 500
  reset
  save all
  op
  let vos = v(vo) - v(vp)
  let vsd9 = v(vdd) - v(vf1)
  let vsd10 = v(vdd) - v(vf2)
  let vsd3 = v(vf1) - v(vmir)
  let vsd4 = v(vf2) - v(vo)
  let vds1 = v(vf1) - v(vtail)
  let vds11 = v(vtail)
  let ptot = -3.3*i(V2)
  set s_vos = $&vos
  set s_vsd9 = $&vsd9
  set s_vsd10 = $&vsd10
  set s_vsd3 = $&vsd3
  set s_vsd4 = $&vsd4
  set s_vds1 = $&vds1
  set s_vds11 = $&vds11
  set s_ptot = $&ptot
  ac dec 50 100 10g
  meas ac a0 max vdb(vo)
  meas ac ft when vdb(vo)=0
  let avec = -1*v(vo)
  let pha = ph(avec)
  meas ac ph find pha when vdb(vo)=0
  let pm = 180 + ph
  * -180 deg crossing of A = where imag(A) rises through zero (no phase-wrap issues)
  let imvo = imag(avec)
  meas ac f180 when imvo=0 rise=1
  meas ac g180 find vdb(vo) when imvo=0 rise=1
  let gm = -g180
  echo "$s_vos $s_vsd9 $s_vsd10 $s_vsd3 $s_vsd4 $s_vds1 $s_vds11 $s_ptot $&a0 $&ft $&pm $&gm" >>
       fc_mc_11_results.txt
  destroy all
```

```
end
quit
.endc
.end
```